Charles Darwin University

PRT661 Data Science Practice Unit

Assessment 1

Project Proposal and Design

Solar Generation Forecasting for Battery Dispatch

DKASC Alice Springs · 2008–2026

Theme 2: Predictive Analytics and Forecasting

Group 4

Member	Student ID
Uttam Shrestha	s396560
Abhishek Tamang	S396739
Sarin Uprety	S394012
Yogesh Basnet	S394043

Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: SCRUM — PRT661 Solar Forecasting Group 4

Submitted 16 August 2026

1. Project overview and objectives

Group 4 will predict photovoltaic generation at the Desert Knowledge Australia Solar Centre (DKASC) site in Alice Springs, 1 to 24 hours in advance, to inform a site owner or operator on when a battery should charge or discharge. Forecast and uncertainty are the deliverable, it's not a system that we build, but the scheduling of the battery is the operational use it informs. Four goals are identified: create a batch pipeline that can be reproduced for nineteen yearly extracts (2008-2026) containing some 1.9 million rows and 3.1 GB of data; develop degradation-aware and ramp-sensitive features; evaluate proposed models against a published benchmark for this site; and display forecasts and forecast errors in a dashboard. It has a design that covers all five of the capabilities: acquisition, storage, processing, analytics/machine learning and visualisation.

2. Theme justification

The project is located in Theme 2, Predictive Analytics and Forecasting, which predicts a continuous future outcome based on historical data, with time-series forecasting being explicitly mentioned in the examples in the theme. Theme 1 was rejected, since dispatch decision does not need to be done with the minimum latency, so it is a batch pipeline and not a streaming one. The reason for excluding Theme 3 is because per-inverter anomaly channels are out of scope, and the fifty or so per-array columns are not included in the feature set for this reason.

3. Problem statement

Solar generation at Alice Springs is dependent on the presence of cloud and on the season, plus a slow degradation of the solar panels as they age. A battery only pays for itself if the operator can predict what generation will be available, and this battery energy storage system was installed in 2024 at the site. Thuseethan et al. (2025) predict this same location, but accuracy is an objective, they use 2019 to 2024, and they pre-date the battery. Three gaps follow. The 2024 installation provides dispatchable SoC (State of Charge) signals. The 17-year period shows degradation not apparent in a 5-year period. The previous studies reported high error at dawn/dusk and peak irradiance but did not investigate what aspects cause the high error. This project is designed to forecast to support operational decision and directly tackles ramp events and feature importance. We are going to use their best published result - R^2 0.8641 and MSE 22.41 on Master Meter 2 to measure up against.

4. Initial architecture

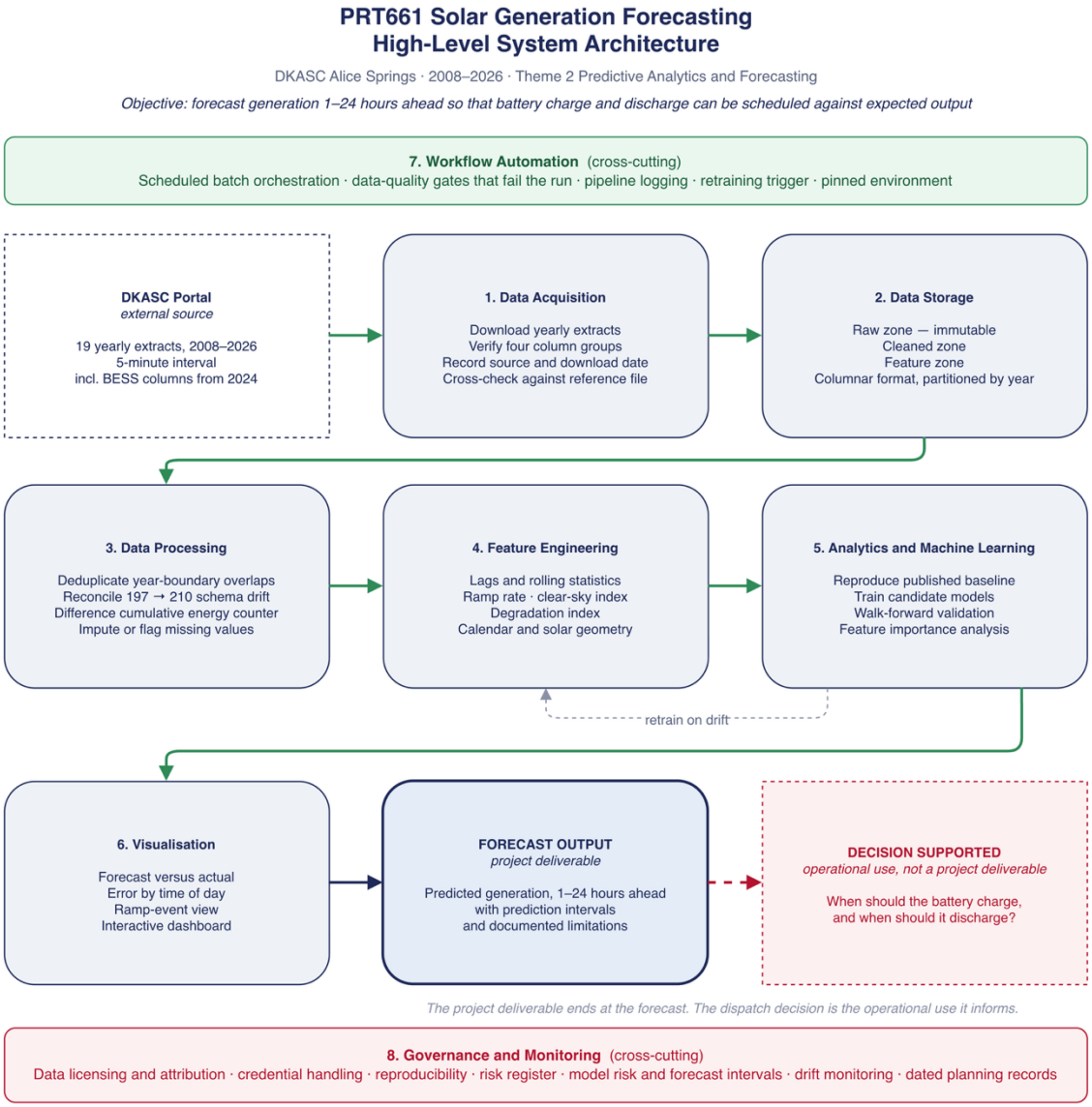


Figure 1. High-level system architecture. The forecast output is computed independently of the dispatch decision that it provides.

The high-level design is shown in Figure 1. The data follows six stages from the DKASC portal to a forecast, with workflow automation and governance spanning all six stages. The forecast is distinct from the decision, which is the edge of what this project will provide. The storage uses parquet files queried through DuckDB; PostgreSQL was rejected as at 1.9 million rows, the overhead would not be justified, and Apache Spark as the working set would be in memory. These both add to the cost of operation, but do not enhance the results. All components are open source and can be reproduced from the repository and Figure 2 shows the design, layer-by-layer.

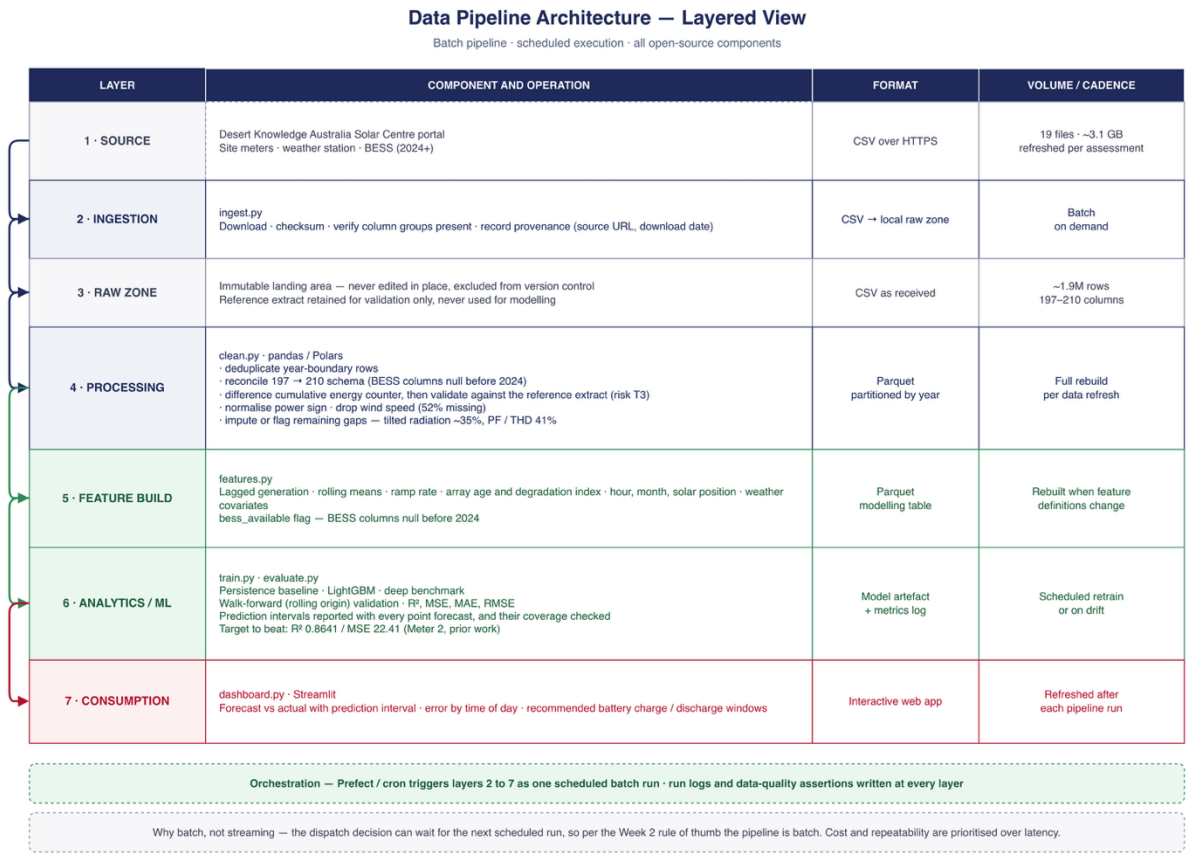


Figure 2. Data pipeline architecture layered view: component, output format and rebuild cadence at each layer.

5. Workflow plan



Figure 3. Workflow plan: the eight epics against the four assessment milestones.

The following eight epics are shown against the four assessment milestones in figure 3. The eight names of epic in figure 3 are the eight epic words on the Jira board, and they are worded exactly the same, and each epic is assigned to its accountable owner, as a Jira assignee. The board has 18 tasks,

each of which has been assigned to a member of the team and to a sprint, seven of which are Done, four In Progress and seven To Do, across four sprints corresponding to the A1, A2, A3 and A4 work areas. Progress, scope changes and completion status are recorded as work takes place and not regenerated when it is submitted.

6. Task allocation and team structure

Group 4 has four members. Table 1 provides a list of roles and epic ownership: each member owns two epics across the entire pipeline order with the Project Lead also having named tasks within Feature Engineering and Analytics and Machine Learning as second member, ensuring that each epic has two members who are familiar with it. Work is done when it is committed, reviewed by another member and their Jira task closed with an evidence link. Decisions are made documented on dated planning records and not agreed verbally.

Role	Member	Epics (accountable)	Report sections
Project Lead	Uttam Shrestha	Workflow Automation, Governance and Monitoring	5, 6, 9 and assembly
Data Engineer	Abhishek Tamang	Data Acquisition, Data Storage	1, 3
Data and Feature Engineer	Sarin Uprety	Data Processing, Feature Engineering	4, 7
Modelling and Visualisation Lead	Yogesh Basnet	Analytics and Machine Learning, Visualisation	2, 8

Table 1. Roles, epic ownership and report sections.

7. Initial risk analysis

The repository's risk register lists 19 risks under the categories of technical, project and governance, including likelihood, impact, mitigation and a risk owner. The seven most significant, by impact, are shown in Table 2. The register is checked at each stand up meeting and each submission; review of 12 August resulted in giving owners to named members and thus matching each risk to a named Jira assignee. Just two risks have been realised and mitigated; the first being the slippage against the internal draft deadline and the second being a stale extract which ends 11 February 2026.

ID	Risk	Impact	Mitigation
T1	Schema drift: 197 columns to 2023, 210 from 2024	Medium	The pipeline takes care to reconcile both schemas explicitly, indicating that the BESS columns are only available from 2024 onwards.
T3	Cumulative counter is giving the wrong readings as interval energy.	High	The cumulative series is differenced, checked for validity with the reference extract and unit-tested prior to use.
T4	High missingness: wind speed 52%, tilted radiation ~35%	Medium	Wind speed is removed and the other missing values are imputed or flagged, and it is noted which of these was done.

ID	Risk	Impact	Mitigation
T5	Stale extract: 2026 data ending on 11th February 2026	Medium	Before A2, the data is re-downloaded and the download date is noted; and A1 is truthful about its coverage.
T8	Overfitting through inappropriate validation	High	No random shuffling at any time, instead it is walking forward validation.
P1	Uneven contribution across the group	High	All work has a named Jira assignee, is discussed weekly in the stand-up and issues are escalated upfront rather than on submission.
G4	Poor dispatch decision due to inaccurate forecast.	Medium	Prediction intervals are reported for forecasts and known weak periods are openly reported as limitations.

Table 2. Extract from the initial risk register, rated by impact.

8. Ethics, privacy and security

No personal information is included in the dataset. It simply documents electrical data from a demonstration facility and data from a co-located weather station, and no one can be identified in either case, thus no consent process is necessary. Data here and in repository is credited to DKASC, and source URL and date of download recorded to allow results to be traced back to a specific extract; raw files are not version controlled.

Seasonal calendars of First Nations were used in previous work on this site. We intentionally do not emulate that approach. This is because it is the substantive work of the authors, and Indigenous seasonal knowledge is cultural intellectual property and should only be used following engagement with Traditional Owners, which is beyond the scope and timeframe of this unit.

There is an operational cost to a poor forecast – not just a statistical cost. Prediction intervals are thus reported along with point forecasts, known weak time periods are recorded, and the system is stated to be a decision support system, not an automated controller. Credentials are never committed: .gitignore excludes environment, key and credential files. Where generative AI may be used to aid in drafting or coding, its use is identified in accordance with CDU policy.

9. References

Data download | DKA solar centre. (n.d.). In *dkasolarcentre.com.au*. Retrieved 16 August 2026, from https://dkasolarcentre.com.au/download?location=alice-springs

Thuseethan, S., Gangajaliya, S., Hamlin, L., Shanmugam, B., & Thennadil, S. (2025). Conv-ensemble for solar power prediction with first nations seasonal information. *IEEE Open Journal of the Computer Society*, *6*, 884–895. https://doi.org/10.1109/OJCS.2025.3580339

Project Update Summary

Charles Darwin University, Danala (Darwin City) Campus (Group 4)

PRT661 Data Science Practice

Semester 2, 2026

Members:
Uttam Shrestha (s396560)

Abhishek Tamang (S396739)

Sarin Uprety (S394012)

Yogesh Basnet (S394043)

Changes Since Previous Assessment

Area	Previous State (A1)	Current State (A2)	Justification
Primary dataset	19 yearly extracts, 2008 to 2026	Master Meter 1, schema evidence, yearly files and 2024 telemetry.	There are no weather columns on the yearly extracts. Master Meter 1 does and is the benchmarked series.
Scale and schema drift	~1.9 M rows, 3.1 GB; 197 columns to 2023, 210 from 2024	1,824,033 rows (3.11 GB), Master Meter 1 holds 1,767,903; 406 distinct column names, 14 in every year, 3 empty, 13 new at 2024	All the estimates were in A1 and were now measured from across the 19 files. The new additions for 2024 include BESS, PV and grid telemetry.
Cumulative counter	Difference and unit test (risk T3)	Six digit register wrapping at 1,000,000; 6 wraps corrected	Differencing alone gave minus one million per wrap. Power-integrated energy is repaired to 99.7% total.
Degradation	Degradation-aware features proposed	-0.454 kW per year (-0.94% of mean), fitted from 2014	The site was still being commissioned and was fitting from 2009 returns +0.748 kW per year.
Model specification	Evaluated against a benchmark	Gradient boosting at 1 h and 24 h; Three baselines; MAE, RMSE, R2, MBE; 80% intervals	Closes A1 feedback that these were unspecified.
Jira evidence	Board named, no URL	Direct board URL provided, staff access turned on	A1 feedback: Tasks and assignees could not be verified.

Contribution Summary

Team Member	Contribution
Uttam Shrestha Project Lead	Profiling of all 19 extracts, schema reconciliation, selected Master Meter 1 after no weather columns were found, ingestion and the Parquet store, the eight quality gates, risk register, AI disclosure, sprint planning, report assembly. *notebooks/01, 01b, 02; SCRUM-40 to 43*
Abhishek Tamang Data Engineer	Acquisition of the DKASC extracts and the storage layout, validate the ingestion to the full five-minute grid, integrity of timestamps, pre-cleaning quality baseline. *notebooks/02; SCRUM-28, 31, 32*
Sarin Uprety Data and Feature Engineer	The leakage-safe shifting of the 30 feature hourly store: *notebooks/03, 04; SCRUM-33, 34, 35*; Five stage cleaning at 5 minute resolution; repair of the six digit register wrap, validated to 99.7%.
Yogesh Basnet Modelling and Visualisation Lead	The walk forward protocol, gradient boosting at 1 h and 24 h against three baselines, the training-loss correction, 80% intervals and coverage, benchmark parity, ten figures, *notebooks/04b, 05, SCRUM-36-39.*

Evidence Locations

Evidence	Repository Location
Profiling, schema drift, ingestion, quality baseline, cleaning	notebooks/01, 01b, 02, 03; Outputs/schema_matrix.csv, schema_drift_summary.md, data_quality_ and cleaning_report_mastermeter1.md
Features, figures, models, metrics, intervals, risk, ethics, governance, PM	notebooks(04, 04b, 05): feature-dictionary.md, model_results.md, figures/fig1-10: Supporting_Documents/, Task_Allocation/, Project_Planning_Records/

Assessment 2 Technical Development

1. Scope, and what changed since Assessment 1

Assessment 1 advocated for a batch pipeline over nineteen yearly DKASC extracts; degradation-aware and ramp-sensitive features; comparison to a published benchmark of the same site; and a dashboard. The operational use was the same: predicting PV power generation one to 24 hours in advance to allow a site operator to determine when the battery should be charged, held or discharged.

Profiling the archive before writing the ingestion code caused the design to be changed in one important respect. The nineteen yearly extracts have a total of 1,824,033 rows and 406 distinct column names, but **no meteorological column**. They log electrical data on a per-inverter basis and do not provide any information on the data. A solar forecast based on that archive would involve irradiance-free modelling of generation.

The Master Meter 1 reference extract is downloaded from the same portal, includes 1,767,903 rows of data at the same five-minute resolution and comes with global horizontal radiation, diffuse horizontal radiation, tilted radiation, air temperature, relative humidity, wind speed and direction and daily rainfall in addition to the target. It also is the actual series that Thuseethan et al. (2025) report their results on, making it possible to do benchmark comparison, instead of approximate. Master Meter 1 was then used as the primary dataset. Extracts are created each year and kept as schema evidence, and for use in site-level battery telemetry (available from 2024).

It is a well-documented change of direction and not a decrease of scope. The pipeline, feature families, benchmark and operational framing are all as proposed. Each figure and table listed below may be reproduced from the notebooks listed in the Evidence Locations table.

2. Updated architecture

The layered design in Assessment 1 survives contact with the data. The difference is where the acquisition layer is going to get data from – now, it reads one selected extract, not 19 schemas that drift over time; and where it introduces an explicit profiling phase before acquisition.

Stage	Implementation	Output	Rebuild cadence
Profiling	notebooks/01, 01b. Header and schema scan all 19 extracts	schema_matrix.csv, schema_drift_summary.md	On each new download
Acquisition	notebooks/02: Master Meter 1 extract - timestamp integrity enforced on the whole 5 minute grid	mastermeter1_5min.parquet	On each new download
Storage	Snappy compressed Parquet, columnar, queried using DuckDB	5 minute and hourly stores	Rebuilt from raw
Processing	notebooks/03. Removal of sentinels, range of physical operations, repair of registers, bounded gap interpolation	mastermeter1_5min_clean.parquet, hourly_clean.parquet	Rebuilt from raw
Feature engineering	notebooks/04. Solar geometry and clear sky index, calendar, lags, rolling statistics and degradation.	features_hourly.parquet	Rebuilt from clean
Analytics and ML	notebooks/05. Gradient boosting / three baselines / walk forward validation / quantile intervals	model_results.md, figures 8 to 11	Per evaluation run
Visualisation	notebooks/04b, 05. Ten figures discussing data quality, data exploration and model behavior	Outputs/figures/	Per run
Governance	Cross-cutting. Quality Gates, Risk Register, Evidence links.	Supporting_Documents/	Continuous

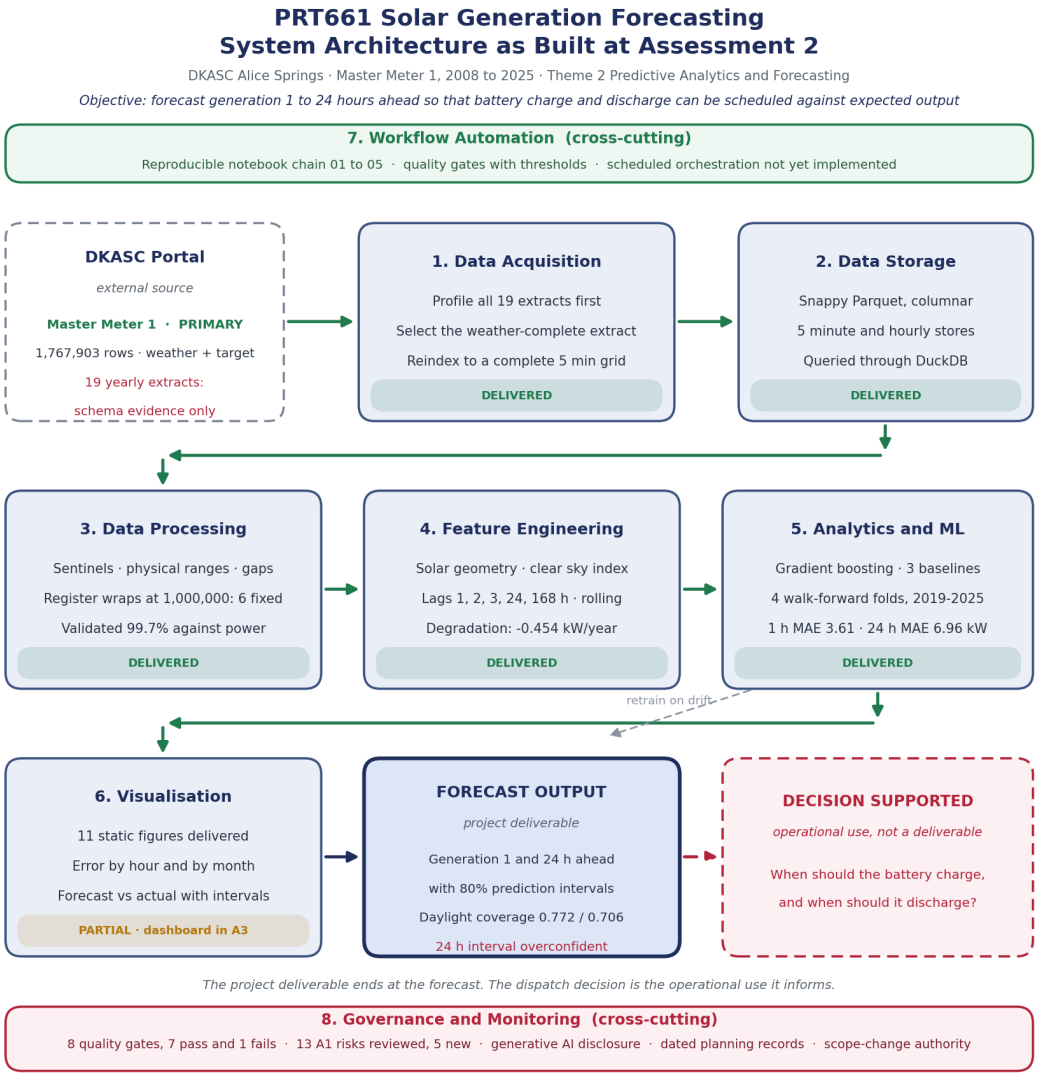


Figure 1. System architecture as built at Assessment 2 (updates Assessment 1's high-level architecture). A delivery marker is present for each component. The schema evidence is provided for the nineteen extracts that are made annually; Master Meter 1 is the primary source. The forecast and the interval of forecast is the project deliverable, the battery charge/hold or discharge decision is outside the system boundary with the operator. The notebook chain has a complementary view committed as `Architecture_Diagrams/06-asbuilt-a2-pipeline.png`.

Two technology decisions that were included in Assessment 1 were revisited and both are maintained. But PostgreSQL has not been justified: the working set is 148,537 hourly rows, which will fit easily in memory, and a server process will add operational costs without any benefit. For the same reason, Apache Spark is not justified. With Parquet with DuckDB, reads are columnar, meaning the modelling notebook only reads the columns it needs from a 31 MB feature store, not re-parsing a 390 MB CSV.

What Assessment 1 wanted to separate is maintained all the way through: The system **delivers a forecast and an interval, not a dispatch command**. Section 9 provides guidance on how an operator can convert one to the other, this is a human decision that is based on the output.

3. The five Vs of this dataset

The five Vs were explored in prose in Assessment 1. As suggested in the Assessment 1 feedback they are tabulated here with measured values.

V	Measured position	Consequence for the design
Volume	3.11 GB across 19 yearly extracts (1,824,033 rows) plus a 390 MB reference extract (1,767,903 rows). After hourly aggregation: 148,537 rows and 31 MB.	Large enough not to be able to use spreadsheet tooling and large enough that storing it as a columnar store is necessary; small enough that using a distributed engine would be pure overhead.
Velocity	Batch. The portal is published periodically, and the current extract will finish on 11 February 2026 for the yearly files and 23 August 2025 for Master Meter 1. No streaming source.	Confirms Theme 2 not Theme 1. The pipeline is not continuously ingesting, instead it rebuilds on download and dispatch is hourly, not sub-second.

Variety	There are 406 different column names in the archive, and 14 columns are present in all years, with 3 of them being empty. 13 columns appear at 2024. One archive for electrical, meteorological and battery telemetry.	Schema reconciliation is not just an afterthought, but rather a first-class pipeline stage. Because of this, it was also used to determine the choice of Master Meter 1, the only extract which had a stable schema and weather.
Veracity	Sentinel codes (999999, -99999), physically impossible values (irradiance to 2,725 W/m2, humidity to 131%, temperature to -40 C), a six-digit register which wraps and missingness from 2.3% on the target to 52.8% on wind speed.	All cleaning rules are based on instrument physics, not on own percentiles of the data and the register repair is tested against an independent estimation.
Value	A 1 hour forecast with MAE 3.61 kW and interval 80% and a 24-hour forecast with MAE 6.96 kW when compared to a site with a range of hourly output of 209 kW.	Enough to guide decisions on battery charging and discharging at intra-day and day-ahead time horizons, with quantified uncertainty at these time horizons.

4. Data acquisition and storage

Profiling was done first, intentionally, so that ingestion would be written against what's in the files, and not what Assessment 1 thought was in the files. The row counts were not based on parsing the files into memory and only the header and three sample rows of each extract were read to construct the column matrix.

The result fixed two issues in the claims that were found in Assessment 1. **1824033 rows are in the archive, not "around 1.9 million" rows**; and the schema is much worse than the reported increase from 197 to 210 columns. A total of 19 files contain 406 different column names. Only 14 are present in each year, with three of those (the archived M15 QCells columns) entirely blank in recent years, with 11 columns of continuous coverage: the timestamp and the M1 array in the three phases. The other 392 columns are inverters that have been commissioned and decommissioned over 17 years of site operation.

The 13 columns that appeared at 2024 were identified as being the site-level telemetry that was added with the battery installation: BESS active, reactive, apparent power, and state of charge; PV power total; grid demand power total; site demand power total. This is a stronger position than Assessment 1 anticipated, as surplus (photovoltaic generation minus site demand) and battery headroom (state of charge) are both directly measured from 2024 onwards. They are only used for Assessment 3, when the dispatch layer is developed.

The quieter finding is for those who are reproducing this work: **every yearly file spills over into the next year one day**. For example, the 2015 extract is from 1 January 2015 to 1 January 2016 at 23:55, which equates to 366 days and 288 readings a day. Just appending the nineteen files together would silently duplicate about 288 rows at 18 interfaces. The problem is avoided by using the single Master Meter 1 extract and the finding is noted against SCRUM-16.

Ingestion's timestamp integrity is maintained by reindexing the data on a full 5-minute grid across the observation period. This turns implicit missingness (when a reading is not made) into explicit missingness. Without it, none of the missingness is evident, and any lag feature is silently filling in an unknown time period. The reindex added 14,519 empty slots to a 1,782,422 slot grid and no duplicate timestamps and no unparseable timestamps.

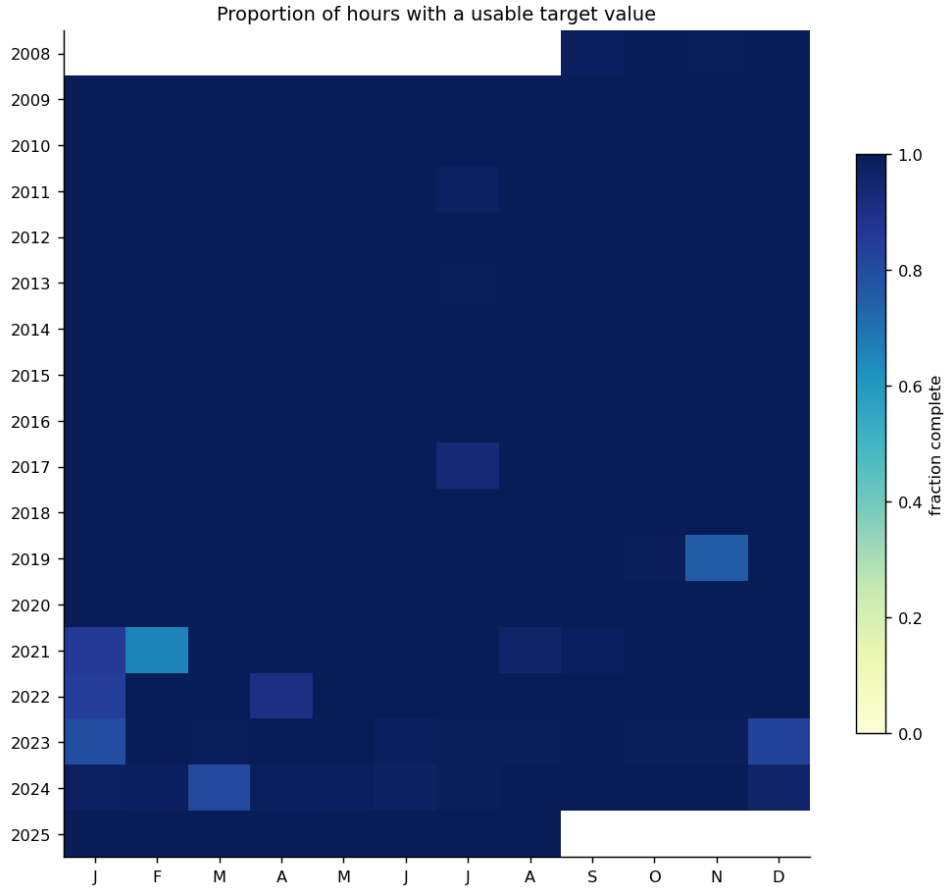


Figure 5. Percent of hours with usable target value, by month. Overall completeness is 99.06%. The weakest months are February 2021 at 65.8% and November 2019 at 75.7%. The white bands at the beginning of 2008 and the end of 2025 represent the range of the extract, not outages.

Storage is Snappy-compressed Parquet at 5 minute and hourly resolution, queried via DuckDB. The .gitignore file excludes raw and derived data, but the small analysis artefacts in Outputs/ are explicitly re-included so that every number in this report can be traced back to a committed file.

5. Cleaning and preprocessing

Cleaning is performed at 5-minute resolution and then the hourly store is rebuilt. Ordering is intentional and was corrected during development: a previous version of the package aggregated first, leading to the propagation of sentinel values to hourly means and a minimum air temperature of -39.988 C in the hourly store.

There are 5 stages that are followed in order.

Stage	Rule	Rationale	Effect
Sentinel removal	Absolute value at or above 9,999 in an instrument channel is considered to be null.	Loggers do not record blanks, but use fixed error codes instead. None of the instruments here have an authentic reading of that magnitude.	Eliminates -99,999 wind directions and tens of thousands tilted radiation
Register diagnosis	The cumulative energy register is not subject to the above and diagnosed separately.	Cumulative kWh counter is not an instrument reading. If the mean load is around 45 kW, it will rack up millions of kWh over 17 years, and an instrument attached to it will surely kill the column.	Identified a six digit register that wraps at 1,000,000, with 6 wraps
Register repair	Wraps are added back to 1,000,000; remaining negatives and jumps >1000 kWh/5 min are set to null.	If all negative differences were to be clipping to zero, it would create intervals of no generation.	Validated: 6,467,319 kWh from the register with 6,488,854 kWh integrated from Active Power, agreement 99.7%
Physical ranges	Per column bounds from instrument limits and Alice Springs climate record.	Outlier detection based on percentiles allows a systematic fault to determine what is normal. Extremes of temperature recorded locally are approximately -7.5 C and 45.2 C.	The maximum humidity was 100%, and the tilted diffuse radiation ranged between 97,322 and 767 W/m2.
Bounded	Time weighted for runs of up	A 5 minute period is a safe gap to	281,355 values interpolated,

interpolation	to 3 consecutive slots (15 minutes). Target is not interpolated.	bridge; a 6-hour period is not. When a model is filled with the variable to be predicted, agreement is manufactured between the model and the data.	target remains at measured 2.34% missing.

Then at least 6 out of 12 readings need to be taken in an hour or the hour is nulled. An hourly mean over two readings of survival and an hourly mean over 12 readings of survival are not comparable and putting them in the training set together is unspoken corruption of the training set. It produces a store with 148,537 hourly rows of which 147,134 have a viable target.

One interpretation needs stating explicitly, because if one drew the implications from the data without careful thought one would get it wrong. About 50% of the Active Power readings are negative and this is not a fault. The meter is signed: during the night the site has a small standby load, which means that the night time values are around -0.5kW whilst the daytime values are around 209kW. The same applies about the energy register: the steps recorded overnight are real consumption, not corruption, as the energy register records net energy delivered less received. That's about 743,880 such steps and it would be an understatement to say that this is corrupt as it is 43% of the record.

Absence is irreparable in a Cleaning. Wind speed is still missing at around 52.8%, power factor at around 41.7%, current phase average and total harmonic distortion at around 41.7%, and the two tilted radiation channels at 34% to 37%. These were never recorded rather than recorded wrongly. They are excluded from the feature set on completeness grounds, and this confirms the exclusion of wind speed that was anticipated in Assessment 1 as risk T4.

6. Feature engineering

Thirty features are built in five families. A fundamental limitation is that no value at a point in time t can be based on any information recorded at or after t (i.e., every lag and rolling statistic is backwards shifted, and the contemporaneous target is not an input to its prediction). This is the basis for the walk forward results in Section 8 to be relevant instead of self-confirming.

Family	Features	Why
Solar geometry	solar_elevation, ghi_clearsky, is_daylight	Elevation from declination, equation of time, hour angle; clear sky irradiance of Haurwitz model. Physics, not learned parameters, which means that they require no training data and extrapolate properly to new years.
Cloud	clearsky_index and its lags and 24 hour rolling mean	Measured irradiance divided by modelled clear sky irradiance. It removes the day/night and seasonal fluctuation, leaving only the cloud signal part, the real unpredictable part. Modeling is done natively at undefined night by construction.
Calendar	hour_sin, hour_cos, doy_sin, doy_cos	Cyclical encodings so that hour 23 sits adjacent to hour 0 rather than 23 units away.
Lag and rolling	target_lag{1,2,3,24,168}, ghi_lag{1,2,24}, csi_lag{1,2,24}, target_roll{3,24,168}, csi_roll24, target_std24	Figure 5 is not asserting lag choice, but merely illustrating the partial autocorrelation of target_std24, which has recent volatility as this is what causes an interval to widen on unsettled days.
Degradation	years_since_start	Illustrates ageing of panels over 17 years as captured in Figure 2.

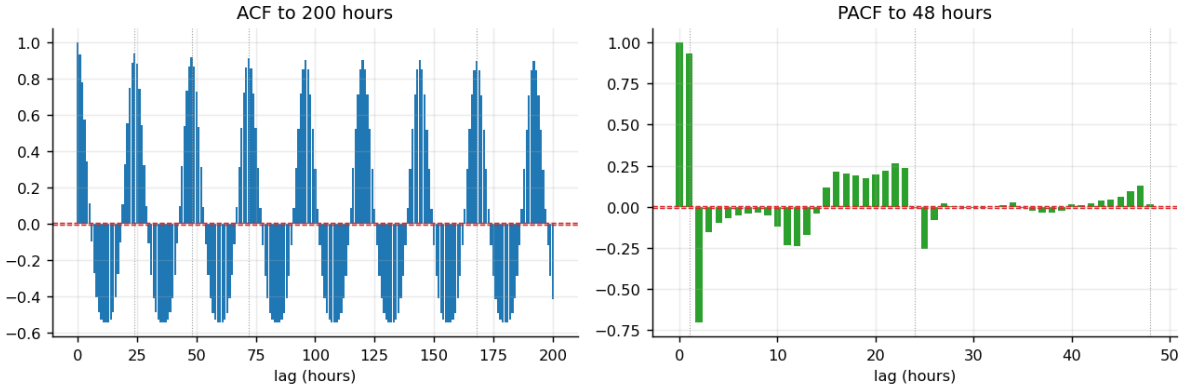


Figure 6. An AutoCorrelation and Partial AutoCorrelation of Active Power computed on the complete hourly grid with time aware shift; thus, lag k is exactly k hours. The ACF peaks at multiples of 24 confirm the presence of the daily cycle in the data, and the PACF indicates which shorter lags contain information not already included in the other lags and justifies the use of the 24 and 168 hour lags.

A previous version of this analysis correlated by position in a daylight series only. There are about twelve daylight hours for each position in the series, and position 24 is about two days, so the axis labelled hours was not hours. Excluding night

is proper when cross correlating with feature(s), inflating the relationship because they are both zero overnight, and inappropriate for autocorrelation, where the daily cycle is the signal. The difference is noteworthy and interesting to note as the wrong one was looking very much like the right one.

7. Exploratory analysis

Six figures were created, each of which is only there to support one of the points this report argues, not to embellish it. Three have results that altered the analysis.

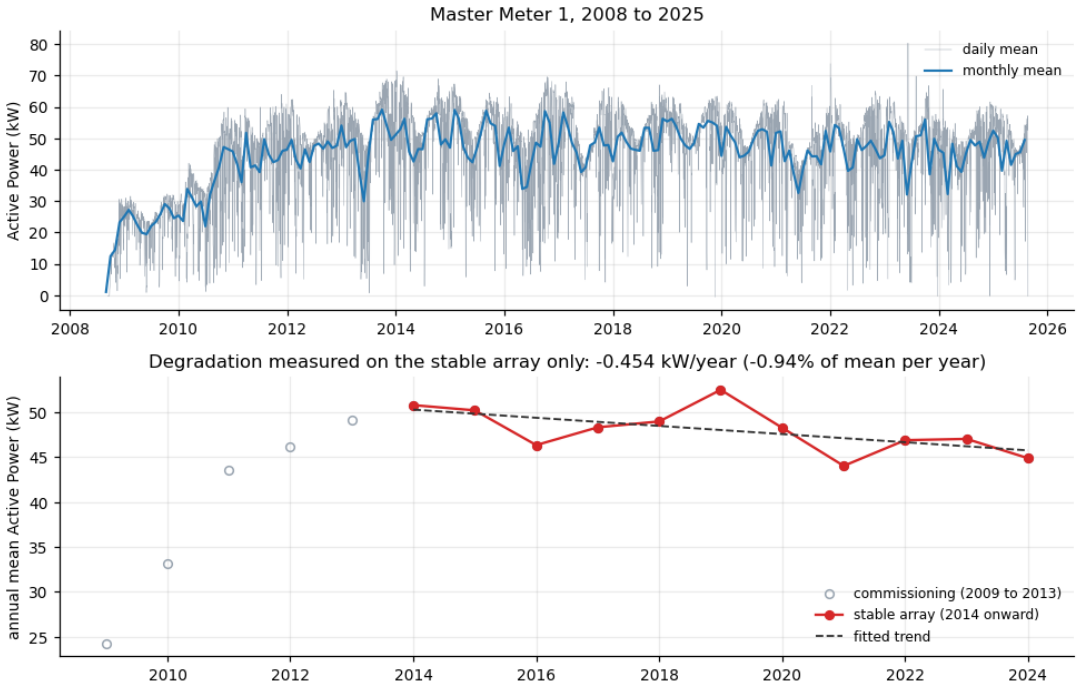


Figure 2. 17 years of Master Meter 1 production. Upper: Daily mean (grey) and monthly mean (blue). Lower: annual means (open markers indicate 2009–2013 commissioning period; only fit from 2014).

The degradation rate is -0.454kW/year, which corresponds to a percentage loss of the mean of -0.94% per year. It is the windowing that is most important, not the number. The 2009 to 2024 data fits at +0.748 kW per year, due to the fact that the site was still in commissioning in the early years, and the output has also increased as arrays were added on rather than falling as the panels age. A degradation chart that indicated generation increment would have been a clear mistake in a degradation aware report. A figure in the stable array, confined to the band of 0.5% to 1% that is generally reported for crystalline silicon, was obtained from seventeen years' worth of site data without any tuning.

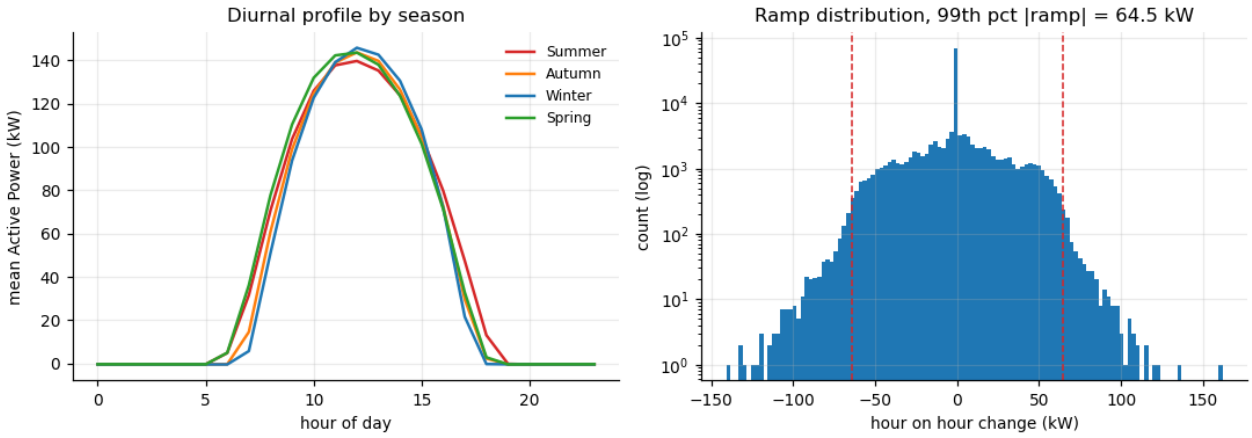


Figure 3. Left: The mean diurnal profile for each season. Right: Distribution of hour on hour change, log scaled, with 99th percentile of absolute ramp at 64.5 kW marked.

There are two characteristics of the diurnal panel that appear to be incorrect but are not. This is marginally higher during the winter than the summer (around 143 kW versus 140 kW) due to Alice Springs being almost on the Tropic of Capricorn (which means that the winter sun is still high in the sky); the clearer winter skies; and the cooler temperatures which make the panels more efficient. The summer is more spread out than intense: the seasonal energy difference is in the length of the day not the intensity. That has a direct impact on operations, because a battery can absorb surplus in the summer more hours.

The ramp panel measures the problem Assessment 1 is trying to solve. There are 1,470 hours whose change exceeds 64.5 kW, with extremes of +162.1 kW and -140.8 kW. With a site with a peak near 209kW, one hour can elapse more than two thirds of full output.

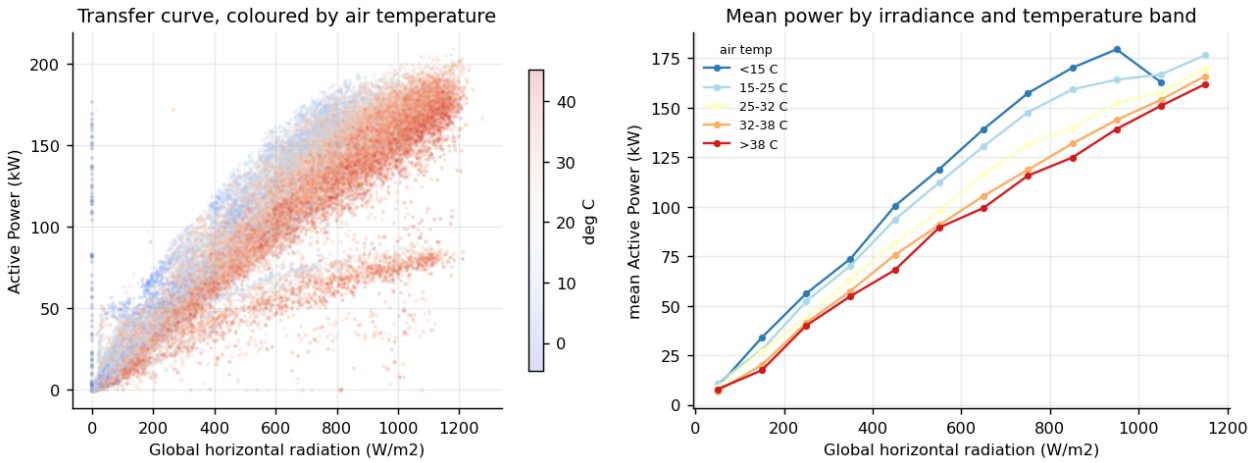


Figure 7. Left the physical transfer curve in color according to the temperature of the air. Right: Irradiance band averaged mean power as a function of temperature band.

The irradiance and the power are correlated with r = 0.903 during daylight hours. The right panel shows the temperature effect: at 700 to 800 W/m2, mean output falls from 157.3 kW below 15 C to 115.7 kW above 38 C, a drop of 26.5%. This is approximately twice that temperature alone can account for. Typical silicon coefficient is about -0.4% per degree over 29 degrees, which would give ~12%. The rest is an angle of incidence confound: a cold hour at 750 W/m2 is typically a winter midday when the sun is high in the sky, and a hot hour at the same horizontal irradiance is more likely to be a summer late afternoon when the sun is at a low angle to the panels. To attach the entire derating (26.5%) to temperature would be an overstatement, therefore the derating is a combined one, and Assessment 3 will separate the two.

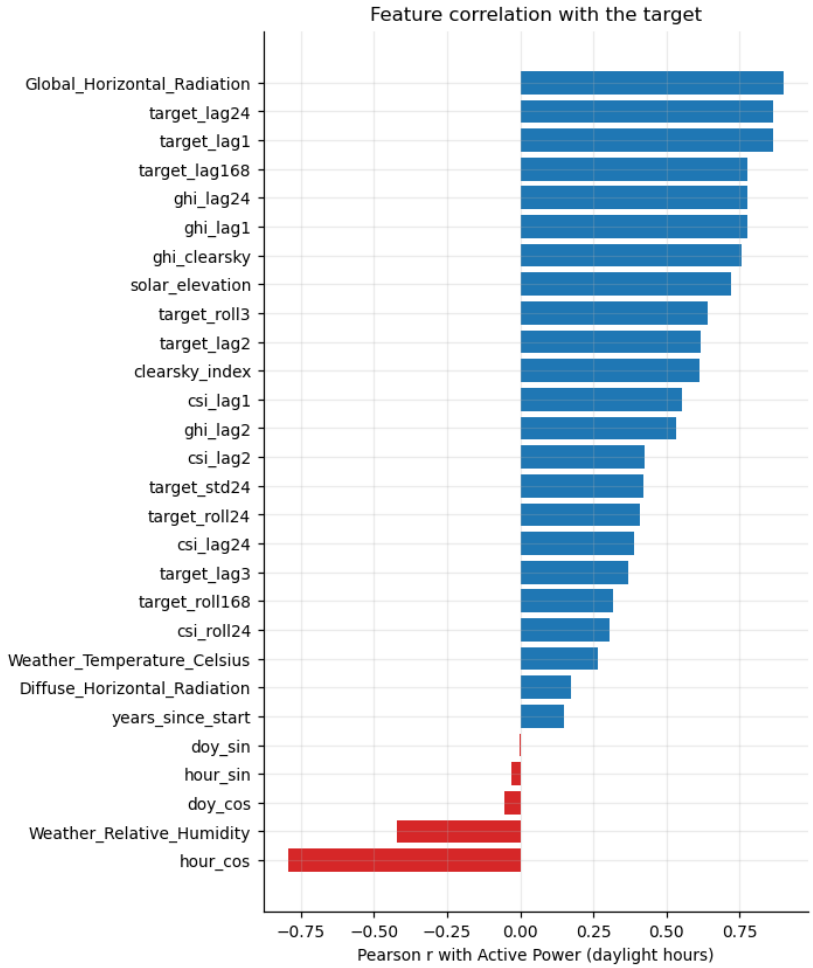


Figure 4. Pearson correlation for all features and the target (day light hours only). The night is omitted because there is no information to be gained from a correlation that has a common zero at night.

The correlation of features with the target over daylight hours (Figure 4) shows that GHR has the highest correlation (0.900) followed by the correlation values of 24-hour target lag (0.863) and 1 hour target lag (0.862). Unfortunately, the clear sky index is not as high as would be desired, at 0.613, meaning that the raw irradiance contains more direct information than the normalized form, despite the latter's theoretical advantages. Both are kept; it is up to the model to decide.

8. Analytics and machine learning

Feedback in Assessment 1 noted that the modelling plan failed to include the candidate models, baselines, horizons, error measures, prediction interval method and model selection criterion. There are six, all of which are detailed here and all of which are in place.

Element	Specification
Horizons	1 hour and 24 hours in advance. The features are measured at time t, and the label is measured at t+h, therefore there is no assumption of weather forecast and no feature value is measured after the forecast time.
Candidate model	HistGradientBoostingRegressor. It was selected since it can naturally accommodate missing values as occurs when the clear sky index is not defined at nighttime, by construction, and because it can model the non-linearity of the sunrise and sunset phenomena that a linear model cannot.
Baselines	Persistence, prediction = y(t). Seasonal naive, prediction = y(t+h-24), the same clock hour yesterday. The mean for that month and hour for climatology. The computation of climatology is performed using only the training slice and not the entire series.
Error metrics	MAE and RMSE in kilowatts, R2, and mean bias error to identify systematic over and/or under forecasting. Skill score is 1 - MAE_model / MAE_persistence.
Prediction intervals	The 10th and 90th percentiles of quantile regression were used to determine the nominal 80% interval, coverage (PICP) was also assessed as well as mean width.
Validation	Four expanding-window walk forward folds with the period of time running from 2019 to 2025. Training is always done before testing and data is never shuffled.
Selection rule	Lowest mean MAE over all the folds, standard deviation across the folds reported. If a model is unstable over folds but it wins on the mean, it will not be selected.

Here is one methodological comment: one of the reasons the conclusion was altered was because of the method used. The model should be trained on the loss that it is evaluated by. By default, gradient booster uses squared error. The 24 hour was trained that way and reported an MAE skill score of -0.219 (lower than persistence) and an unstable fold 3 at MAE 13.95 (versus 7.0 to 8.8 elsewhere). When retrained with absolute error loss, both were reversed: skill became +0.101 and the fold standard deviation dropped from 3.096 to 0.800. The 2023 data problem was an artefact of the loss function and NOT of the extract itself..

Model	1 h MAE	1 h RMSE	1 h R2	24 h MAE	24 h RMSE	24 h R2
Gradient boosting (MAE loss)	3.658	9.324	0.978	**6.963**	18.552	0.913
Gradient boosting (squared error)	**3.605**	8.948	0.980	9.440	19.537	0.903
Persistence	13.904	23.241	0.866	7.742	21.213	0.887
Seasonal naive	7.744	21.212	0.887	7.742	21.213	0.887
Climatology	11.073	21.157	0.889	11.065	21.148	0.889

Table: Average over 4 Walk Forward Folds. The best MAE per horizon is in bold. By construction, the seasonal naive baseline is the same as persistence at the 24 hour horizon, where y(t+h-24) is the same as y(t).

The model at one hour achieves MAE of 3.61 kW and R2 of 0.980 with a skill score of +0.74 compared to persistence. The performance of persistence is poor, at hourly resolution the diurnal cycle dominates the hour-to-hour change, which explains why it is outperformed by seasonal naive at 7.74 kW, nonetheless, even given this stronger baseline, the model still outperforms by 53% in terms of MAE.

As of 24 hours, the honest position is smaller. At one hour, MAE 6.96 kW is 10% better than the best baseline of 7.74 kW, which is materially smaller than the above. It is not because it is a failure of the model, but because the structure of the situation is as follows: If there is no modelling from the NWPs at all, then the best information about tomorrow at noon is what happened today at noon, and a model can provide only a limited improvement over that. Most clearly this is the case in relation to the Assessment 3 work detailed in Section 14.

8.1 Comparison with the published benchmark

Thuseethan et al. (2025) report R2 0.8015 and MSE 96.13 for Master Meter 1 at this site. As this project is based on the same series, it can be compared. It is reported with parity conditions, not in a headline, since the two studies do not address the same issue.

Setting	R2	MSE
Thuseethan et al. (2025), contemporaneous mapping	0.8015	96.13

This project, 1 hour ahead forecast	0.9774	87.17
This project, 24 hour ahead forecast	0.9175	317.60
Persistence, 1 hour	0.8610	535.09
Persistence, 24 hours	0.8857	440.53

Four conditions are different and all are mentioned because if these are not mentioned it becomes flattering and meaningless.

Condition	Thuseethan et al. (2025)	This project	Matched?
Target series	Master Meter 1, DKASC Alice Springs	Master Meter 1, DKASC Alice Springs	Yes
Problem framing	Contemporaneous mapping: irradiance at the predicted instant is an input	Predicting ahead: any value after or equal to the forecast time is ignored	**No**
Evaluation period	2019 to 2024	In 2019–25, the number of walkers traveling forward was 110,500.	Partly
Sampling interval and target definition	No data to support setting.	Mean Active Power in kW/ hour	Unknown

Material row is the second row. A nowcast, with the irradiance level at the moment it is forecasting, is solving a simpler problem than a forecast which has to predict it. Our R2s are superior to theirs, but that does not prove that we have a superior method and to suggest otherwise would be the least difficult part of this report to deflate. The defensible statement is more limited: at one hour, the forecast without any future information has an R2 of 0.9774, and at twenty-four hours the value is 0.9175; the published nowcast is a good indicator of the size of the problem, but not a competitor to beat.

Full parity will be achieved by the sampling interval and target definition for the paper and a re-run from 2019 to 2024. It's an Assessment 3 task that's defined.

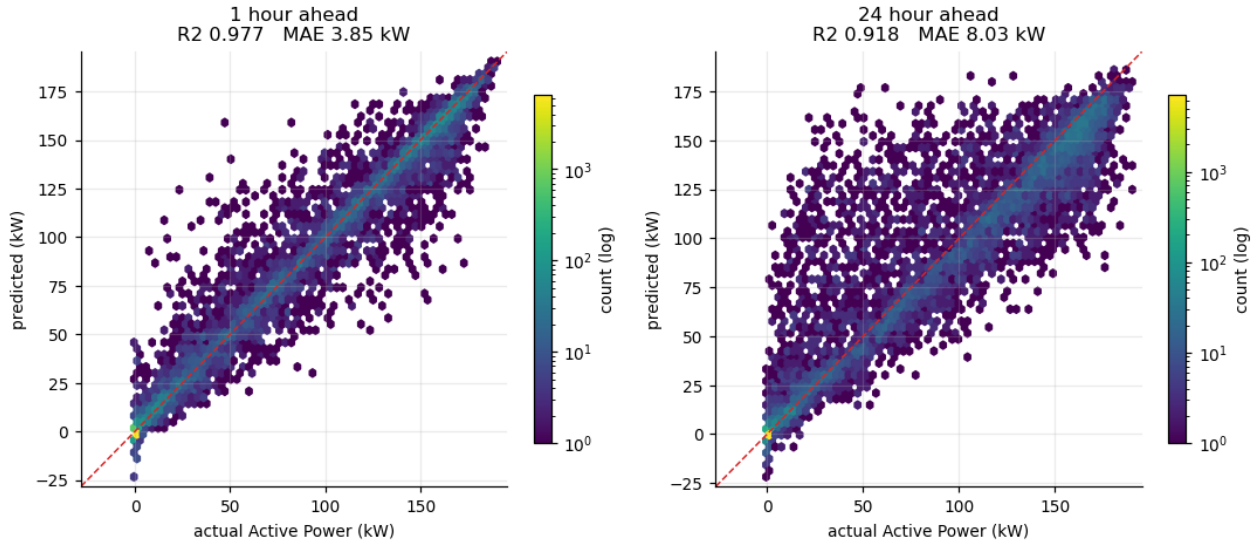


Figure 8. Log-scaled density plotted against predicted. 1 hour model sticks very close to the diagonal. The 24 hour model features the characteristic fan of a forecast without future weather, widest in the mid range where cloud determines the forecast.

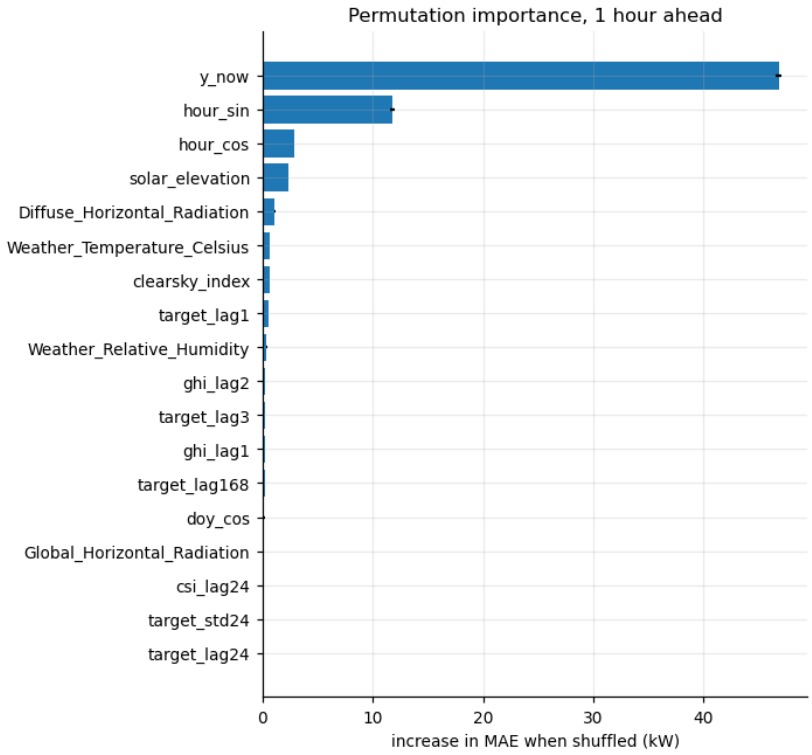


Figure 10. The permutation importance at 1 hour horizon (how much MAE is increased when a feature is shuffled). This is unlike split counts as it shows the impact on the held out performance.

The most important factors are the current observation (46.83 kW increase in MAE when shuffled), hour of day encoding (11.77) and solar elevation (2.33). The sequence is right for a one hour forecast, and a warning: the models at this range are a sophisticated persistence correction, and the weather features deserve to be included primarily at the longer range.

9. Uncertainty, and turning a forecast into a dispatch decision

Assessment 1 feedback focused on how to read a prediction interval and how different horizons would support different decisions, in the context of charging, holding or discharging. Both are covered here.

Horizon	Decision it supports	How the interval is used
1 hour	Suspension/Arrest. The question of whether to start uptake of an imminent surplus or to maintain charges in case generation drops.	The interval is narrow during the day light (mean interval width 24.0 kW). An operator can agree to a charging rate that is close to the minimum and be sure to achieve it.
24 hours	This interval is about twice as large (mean 45.9 kW). It is not a set point; it is a scheduling figure and the lower is the support.	The basic rule of thumb is that in some circumstances when under preparing is cheap such as in the case of charging, the operator gambles on the lower end of the range, and when it is expensive such as in the case of cutting, the upper end of the range is the gambit.

The basic rule of thumb is that the operator "gambles" on the lower bound of the range when the cost of under-preparing is low, for instance in the event of charging, and on the upper bound when the cost of under-preparing is high, for instance in the event of curtailing a large surplus. The point forecast is the case that is expected NOT planning case.

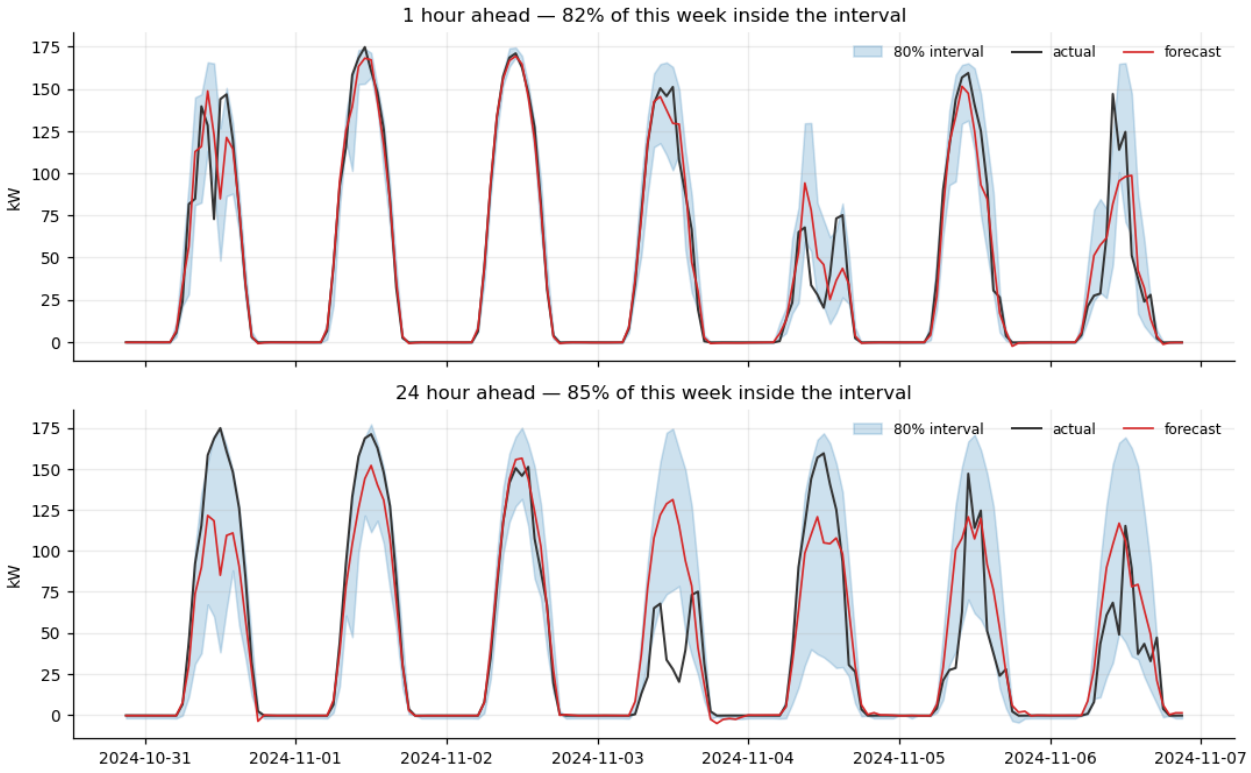


Figure 9A single week of the final fold at the 80% interval shaded. Clear days are tightly forecast at both horizons. On the rough days of 3rd and 4th November the interval is expanded such that it is easily seen by the operator.

Interval quality must be reported honestly and is imperfect. The day coverage is 0.772 at one hour and 0.706 at twenty-four hours, compared with a nominal coverage of 0.80. The one-hour interval is close to calibrated and the twenty four hour interval is overconfident, meaning an operator would be fooled about one out of ten times. It is easy to widen and will be done in Assessment 3.

Coverage by all hours is even lower at 0.702 and 0.691 – but this is misleading and should not be quoted. The interval closes up to a mean width of 1.24 kW and 0.70 kW at night since generation is deterministically very low at night, and a very small wobble will be outside a very small interval. Daylight coverage is the operationally meaningful measure.

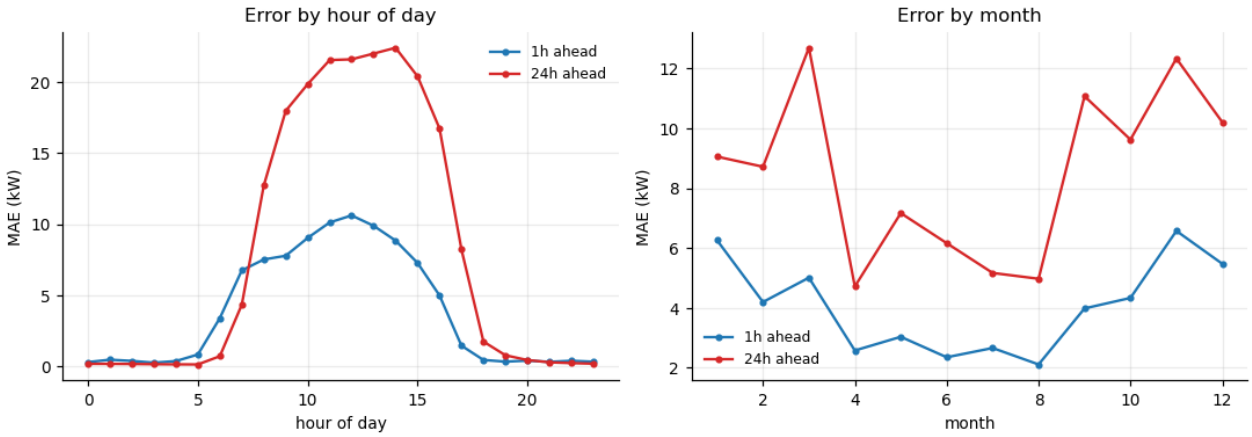


Figure 11. Mean absolute error for each hour of day and each month. All the error is almost totally concentrated in the daylight: 6.77 kW against 0.40 kW at night for the 1 hour model, 14.62 kW against 0.26 kW at 24 hours.

This profile will tell an operator exactly when to trust the number. The overnight forecast is directly correct and has no gap. The error is material between 07:00 and 17:00 and the space between is a real information. The error is highest seasonally in the November to March period when the convective cloud is less predictable due to the influence of the monsoon and lowest seasonally during the clear dry season.

10. Implementation progress and updated workflow

The 8 workflow epics as mentioned in Assessment 1 stay the same and are used in the same way as Jira epics, thus progress can be read directly from the board at uttamshrestha1618.atlassian.net/jira/software/projects/SCRUM/boards/1. Sprint 2 is scheduled to occur in August and has 13 tasks that have a severity of A2 mapped to the epics, with the goal "Build the data pipeline and deliver the A2 progress report".

Epic	Assessment 2 state	Evidence
Data Acquisition	Complete. Primary data set selected on measured bases; schema reconciliation performed for all 19 extracts profiled.	notebooks/01, 01b; SCRUM-40
Data Storage	Complete. Two resolutions for parquet, quality baseline published.	notebooks/02; SCRUM-31, 32
Data Processing	Complete. Five stage cleaning with measured before and after table; independently validated repair register.	notebooks/03; SCRUM-33, 34
Feature Engineering	Complete. 30 features, 5 families, published dictionary and no temporal leakage.	notebooks/04; SCRUM-35
Analytics and ML	Complete for A2. Two horizons, three baselines, walk forward validation, quantile intervals, benchmark comparison.	notebooks/05; SCRUM-37, 38, 39
Visualisation	Partial. Delivered 10 static figures. The interactive dashboard continues to be an Assessment 3 deliverable.	notebooks/04b; SCRUM-36, 22 deferred
Workflow Automation	Partial. The notebook chain is still being reproduced in order; Orchestration in the pipeline is not yet scheduled; the architecture diagram is updated to the "as-built" pipeline.	Figure 1; SCRUM-41; SCRUM-23 carried forward
Governance and Monitoring	Complete for A2. Customisable quality gates with thresholds, evidence links throughout, risk register updated.	Section 11; SCRUM-42

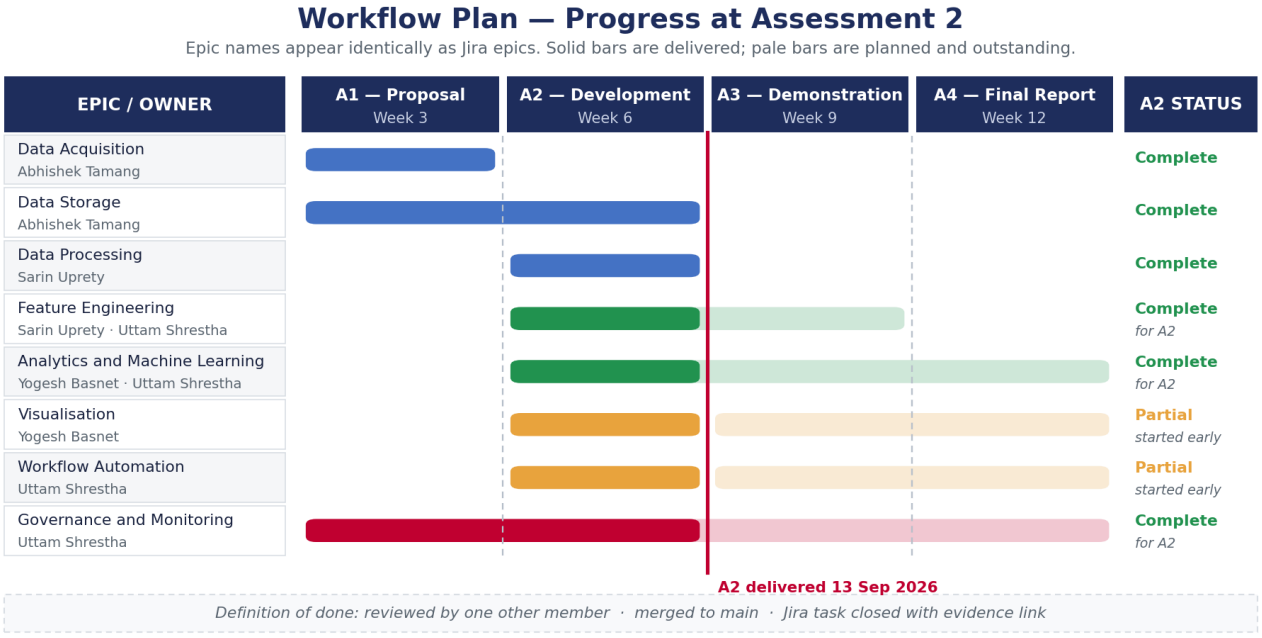


Figure 12. Assessment 1 workflow plan with delivery noted. Solid bars are delivered; pale bars are planned and outstanding. There are 6 complete epics for Assessment 2. The visualisation and Workflow Automation was scheduled to be initiated at Assessment 3 and was started earlier, hence both appear to be in partial delivery ahead of their delivery period.

Pipeline is reproducible from the repository end to end. A reviewer uses the raw extracts on notebooks 01, 01b, 02, 03, 04, 04b and 05, each of which only reads the store generated by the notebook that proceeded it and each generates a markdown artefact in Outputs/ which this report cites. All figures or numbers here are not produced outside the chain.

This is an updated version of both diagrams. The architecture diagram is updated to the pipeline built (Figure 1) and the workflow plan is now delivery on plan (Figure 12). Each is committed alongside the originals, and not over the originals, as the Assessment 1 diagrams record the plans that were designed and erasure of the plans would result in a loss of history of what was proposed versus what was delivered.

Orchestration is the honest gap. The stages execute sequentially, not because of the presence of a scheduler but because a person who is running them does so sequentially. Assessment 1 pledged to workflow automation as an epic and only partially achieved this.

11. Governance, quality gates and acceptance thresholds

Assessment 1 feedback requested measurable acceptance thresholds, a statement of the consequences of a failing gate and a named authority for scope changes. All three follow.

Gate	Threshold	Measured now	Action on failure
Target completeness	A usable target is achieved for at least 95% of hours at any time period evaluated.	99.06% PASS	Stop evaluation, re-download and re-profile before retraining models.
Feature completeness	If the percentage of missing data is > 30%, the feature will not be added to the feature set, and the exclusion will be noted	6 features excluded PASS	Feature dictionary records which and why, exclusion is automatic
Register validation	Energy from repaired register should be within 2% of the energy integrated from the power.	99.7% agreement PASS	Switch back to power integrated series and mark register as unusable.
Forecast performance	Positive skill with respect to the best baseline: 1 h MAE at or below 5.0 kW and 24 h MAE at or below 9.0 kW	3.66 kW and 6.96 kW PASS	Previous version of the model is still used, and regression is investigated; no promotion of model.
Interval coverage	Daylight PICP between 0.75 and 0.85 for an 80% interval (within 5 percentage points of nominal)	0.772 at 1 h PASS; 0.706 at 24 h **FAIL**	Interval published with a clear warning – recalibration planned. Point forecasts continue to be valid.
Fold stability	The standard deviation of MAE over the folds should not exceed 20% of the mean	1 h 10.8% PASS; 24 h 11.5% PASS	Check out the deviant fold first before reporting on a mean.
Pipeline reproducibility	Each printed number and figure can be replenished from the dedicated notebooks.	10 of 10 figures PASS	The artefact is removed from the report until it can be reproduced
Dashboard refresh	Not yet applicable; refresh - within 60 minutes after a new extract is made	Deferred to A3	To be set up with the dashboard

There is a gate that doesn't work. The coverage for 24 hours at 0.706 is outside the range of 0.75-0.85. However, as per the above rule, the interval is not withdrawn, but published with an explicit warning, as an overconfident interval with an explicit warning is more useful to an operator than an interval with no warning, and the point forecast is irrelevant. Recalibration is a named Assessment 3 task.

Authority. The Project Lead approves changes of scope, and any such change is recorded as a dated entry under Project_Planning_Records/ with the reasoning, before the work starts. Technical decisions within the epic is left to the epic's accountable owner. The change of primary dataset mentioned in Section 1 was conducted in accordance with this rule and duly recorded. It is only considered complete when committed, reviewed by another member via pull request and the Jira issue is closed with the evidence link.

12. Risk

Nineteen risks were identified on the Assessment 1 register. Four of these were realized and reported with their outcomes during this assessment and two new risks are added.

ID	Risk	Status	Outcome
T1	Schema drift, 197 columns to 2023 and 210 from 2024	Realised, worse than assessed	The actual drift is 406 different names, and 11 of these are populated for each year. However, this was mitigated by the selection of a single stable extract, instead of reconciling nineteen.
T3	Cumulative counter misread as interval energy	Realised	The register is not monotonically counting but wrapping at 1,000,000. The errors with differencing alone were in the neighborhood of minus one million. Repaired and verified to be accurate to 99.7% compared to an independent estimate.
T4	Weather variability is high: wind - 52%, tilted radiation ~35%	Realised as forecast	Measured at 52.8% and 34% to 37%. The remainder excluded as planned—a threshold was documented, except for wind speed.
T5	Extracts are valid until 11 February 2026.	Realised, still open	Confirmed. Master Meter 1 is even more outdated, coming to an end 23 August 2025. Re-downloaded as SCRUM-14, and carried to current sprint.
T8	Validating improperly to overfit	Mitigated	Four expanding window walk forward folds, with no

			shuffling at any point, all the features are shifted back.
P1	Lack of equal participation in the group	Realised	Three members received an individual contribution deduction for Assessment 1. Mitigated for A2 by file-level ownership, per-member Jira assignment and a documented commit allocation, contribution is verifiable instead of asserted.
T10	*New*. Training loss is not aligned with reported metric	Realised and closed	The squared error training on MAE resulted in a negative skill for 24 h and a bad fold. The two were both resolved with the matching of the loss to the metric.
G5	*New*. Prediction intervals are too narrow at longer forecast horizons	Open	24-hour daylight coverage 0.706 as opposed to the nominal 0.80. Calibration for A3 will be done after publication with a warning.

13. Ethics, privacy, security and use of generative AI

The setting described in Assessment 1 has not changed and is repeated here with what has since been put in place. The data set contains electrical measurements from a demonstration facility as well as weather station measurements from a co-located weather station. None of them reveal a person, do not contain any personal information, and do not require consent. The data is attributed to the Desert Knowledge Australia Solar Centre, and the URL of the source and date of download are noted to enable any result to be tracked back to a particular extraction. Raw files are not version controlled.

Seasonal knowledge of First Nations. This is based on the previous research on this site, which includes First Nation seasonal calendars. This project intentionally does not follow in that footsteps. It is the substantive contribution of those authors and Indigenous seasonal knowledge is cultural intellectual property which cannot be utilised without engagement with Traditional Owners and is outside of the scope and timeframe of this unit. Here the clear sky index and cyclical calendar encodings are based only on solar geometry and the Gregorian calendar.

Running cost instead of a statistical cost. If a forecast is wrong, there are repercussions other than a worse error score. It is not an automatic control system, but a decision supporting system: it gives a forecast and an interval; the decision is made by a person. Every point forecast has a publicly available prediction interval, weak periods are not hidden (Section 9), and a governance gate is recorded as failed and not quietly flipped to a pass.

Security. No credentials are ever committed. Files matching credential patterns and environment files are not included in the repository as are certificate files and key files. This was confirmed in Assessment 2 when the ignore rules were updated to re-enable small analysis artefacts – credential exclusions remained as is and unchanged.

Use of generative AI. Assessment 1 was an agreement to detect any use of Generative AI in line with CDU policy. Here that promise is kept. Generative AI was used in a some way in Assessment 2 in the following ways:

Where used	Nature of the assistance	What the team did
Notebook code	Taking the idea of creating the notebooks for profiling, ingestion, cleaning, feature and modelling as well as the plotting code.	Set out requirements, carried out all notebooks, checked all output and found and fixed a few errors in the code.
Error identification	Checking the output and pointing out discrepancies.	The four substantive errors identified and corrected in this manner were: loss of energy register due to sentinel rule, autocorrelation calculated at incorrect index, degradation trend fitted over commissioning period, training loss not aligned with reported metric. Each is recorded in this report.
Project administration	Checking Jira issue descriptions and the commit allocation.	Project Lead reviewed and applied.

The team did not use AI to generate any results without executing and inspecting them. In the event that AI-generated code was incorrect, the error with its correction is mentioned in the body of this report, and the error is not excised from the record.

14. Evaluation of progress, and Assessment 3

The pipeline is built and reproducible, features are implemented and documented, models are trained and validated at both the stated horizons with quantified uncertainty and benchmark comparison is stated with its parity conditions, compared to the promise of Assessment 1. Assessment 3 tasks still Assessment 3 tasks. Dashboard and scheduled orchestration are not provided.

Four things have improved materially in a better than planned manner. Choosing Master Meter 1 provided the project with a data set that closely matched the weather conditions, but which would have been completely lacking otherwise. The degradation rate of -0.94%/year is not an assumption but a result. The register repair is validated by two independent

derivations agreeing to 0.3%. Not only is the dispatch layer motivated, but also the 2024 site telemetry has been found in the profiling.

There are three things that are not as strong as they should be, and that are not said but left under the surface. The 24 hour prediction interval is too narrow, and one governance gate is broken. Manually handle workflows. The tracking discipline of the project was lost between assessments: The Sprint 1 was left open for a month after the end date and the Assessment 2 sprint was planned in August (with the correct goal) but wasn't begun until the delivery sprint was. The work was not spread out over the period, but delivered in a single intense burst, which will be reflected in the burndown. The team is keeping this on record as a Process Failure and not as a paced delivery.

The single best change you can make is not a better model. It's a more efficient entry.

Assessment 3 priority	Rationale	Owner
Numerical weather prediction input at the 24 hour horizon	It takes a big leap forward from the strongest baseline, and the 24 hour model improves by only 10%, as tomorrow's best is pretty much today's actual. Measuring and supplying forecast irradiance and cloud cover will directly remove the binding constraint and should result in a bigger gain as compared to any algorithm change.	Yogesh Basnet
Recalibrate the prediction intervals	The predicted daylight coverage is 0.706, which is below the stated gate of 0.80. This can be larger quantiles or conformal calibration on a held-out fold.	Yogesh Basnet
2024 onward telemetry:	Build dispatch layer on top of 2024 onward telemetry data that includes BESS state of charge, site demand, PV totals and grid exchange. The surplus and the headroom are no longer hidden but visible and the forecast can be converted into a charge, hold or discharge recommendation.	Uttam Shrestha
Separate temperature derating from angle of incidence	Figure 7 illustrates a 26.5% difference in output at the same horizontal irradiance, of which only ~12% is attributable to temperature. To separate the two effects, and probably improve the feature set, one would want to compute plane-of-array irradiance.	Sarin Uprety
Refresh the extracts and automate the pipeline		Abhishek Tamang
Interactive dashboard	Master Meter 1 will expire on 23rd August 2025 and Master Meter yearly extracts will expire on 11th February 2026. A staleness risk and an orchestration gap would be closed with a scheduled rebuild.	Yogesh Basnet

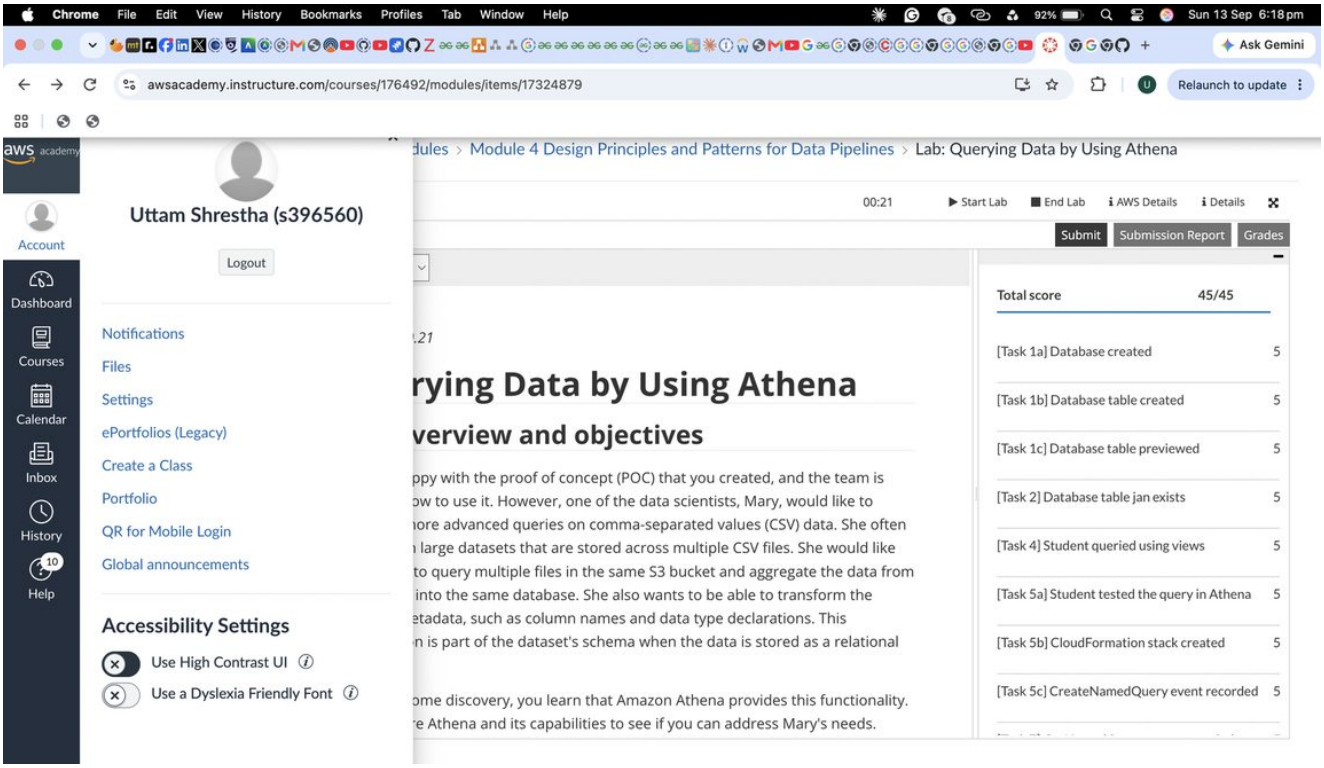
Individual Work and Contribution Summary

Completion evidence: 1 page per student – Laboratory 1 completion evidence, contribution linked to committed artefacts and Jira issues, member's own reflection.

Uttam Shrestha

Student ID s396560 · Project Lead · Epics: Workflow Automation, Governance and Monitoring

Laboratory 1 Completion Evidence



Lab: Querying Data by Using Athena (Module 4). Completed, total score 45/45. Uttam Shrestha (s396560). Captured 13 September 2026, 6:18 pm.

Individual Contribution

Contribution	Evidence
All 19 raw extracts were profiled, and schema reconciliation was performed to correct the number of rows from the raw extract (measured: 1,824,033) and update the number of columns populated in each year,	notebooks/01_data_profiling.ipynb, notebooks/01b_schema_decisions.ipynb, Outputs/schema_matrix.csv, Outputs/schema_drift_summary.md
Noted that the yearly extracts do not contain any meteorological columns and chose Master Meter 1 as the main dataset; noted Section 1 changed direction	Section 1; Task_Allocation/a2-commit-allocation.md
Created the ingestion stage and Parquet storage layer, with the scope changed by the authority rule, and with eight quality gates that provide quantitative thresholds and failure actions	notebooks/02_ingest_mastermeter1.ipynb; Outputs/data_quality_mastermeter1.md; Section 11
Updated the risk register with four realized and two new risks; wrote the generative AI disclosure honoring the Assessment 1 commitment	Supporting_Documents/risk-register.md, ethics-privacy-security.md
Project management and report compilation: A2 sprint planning, 13 issues assigned and scoped among the eight epics, documented commit allocation per member	Jira Sprint 2; Task_Allocation/a2-commit-allocation.md; this submission

Repository: github.com/Uttam1618/PRT661-solar-forecasting · commit history · pull requests · notebooks
Jira board: SCRUM board 1 · **issues:** SCRUM-40, SCRUM-41, SCRUM-42, SCRUM-43, SCRUM-23

Professional Reflection

The best thing I did in this assessment was to profile the data prior to writing any pipeline code. I was hoping to verify what Assessment 1 took for granted and discovered the archive we had designed the entire project around does not have any weather information. We wouldn't have had time to recover if we had written the ingestion layer first and discovered that later.

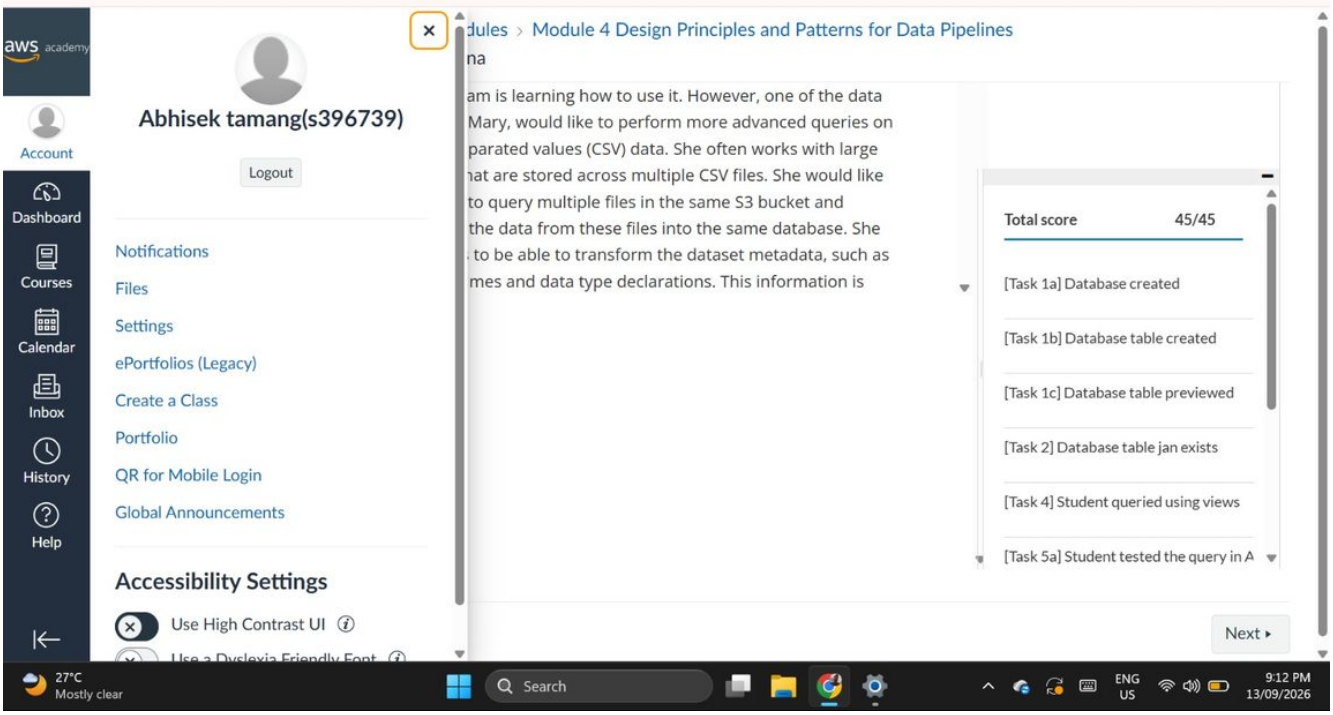
The more challenging one concerns my own presumptions. Assessment 1 had listed a count of rows, and a schema drift figure that was incorrect – no one had measured them. I now regard any number I can't trace back to a committed artefact as a liability and hence the naming of each of these numbers by the notebook from which it came.

I have to say, on process, my project management was documented more than done. This is a sprint that was planned in August and wasn't commenced until delivery date - and I had an open sprint for a month. The disparity is evident in the commit record – it is not evenly distributed throughout the team – and that is a leadership failure first and foremost. My commitment for Assessment 3 is to do a weekly board review as a full team, and to always have a commit from who owns the epic.

Abhishek Tamang

Student ID S396739 · Data Engineer · Epics: Data Acquisition, Data Storage

Laboratory 1 Completion Evidence



Lab: Querying Data by Using Athena (Module 4). Completed, total score 45/45. Abhisek Tamang (s396739), as spelt on the AWS Academy account. Captured 13 September 2026, 9:12 pm.

Individual Contribution

Contribution	Evidence
Staged and downloaded the DKASC annual extracts 08-26 and verified the necessary columns are available on the meters, weather and BESS telemetry	SCRUM-28, SCRUM-13; Datasets/README.md
Developed the raw, cleaned and feature storage format chosen by the pipeline and created the Draw.io	Architecture_Diagrams/; SCRUM-17, SCRUM-18
1,767,903 source rows were reindexed onto a 1,782,422-slot five-minute grid with 14,519 explicit gaps introduced to validate the Master Meter 1 ingestion	notebooks/02_ingest_mastermeter1.ipynb; SCRUM-31
Verified timestamp integrity: zero duplicate timestamps, zero unparseable timestamps for all 17-year period	notebooks/02, section 1 output
Validated data quality baseline measured and used to justify every cleaning decision including per column completeness table	Outputs/data_quality_mastermeter1.md; SCRUM-32
Confirmed extract currency issue. Master meter 1 expires 23 August 2025 and yearly extracts expire 11 February 2026, resulting in a continued risk T5.	Section 12; SCRUM-14

Repository: github.com/Uttam1618/PRT661-solar-forecasting · commit history · pull requests · notebooks
Jira board: SCRUM board 1 · **issues:** SCRUM-31, SCRUM-32, SCRUM-14, SCRUM-28, SCRUM-17, SCRUM-18

Professional Reflection

I believe that in Assessment 1 we were assuming that we would have all we needed in the yearly extract. Data checks were not conducted adequately, in particular the availability of weather data in the files for the project. Later in the day this was an issue because the weather columns were not included in the files, so the Master Meter 1 reference extract was used. It helped me to understand the need to check and understand the data prior to designing the project around the data.

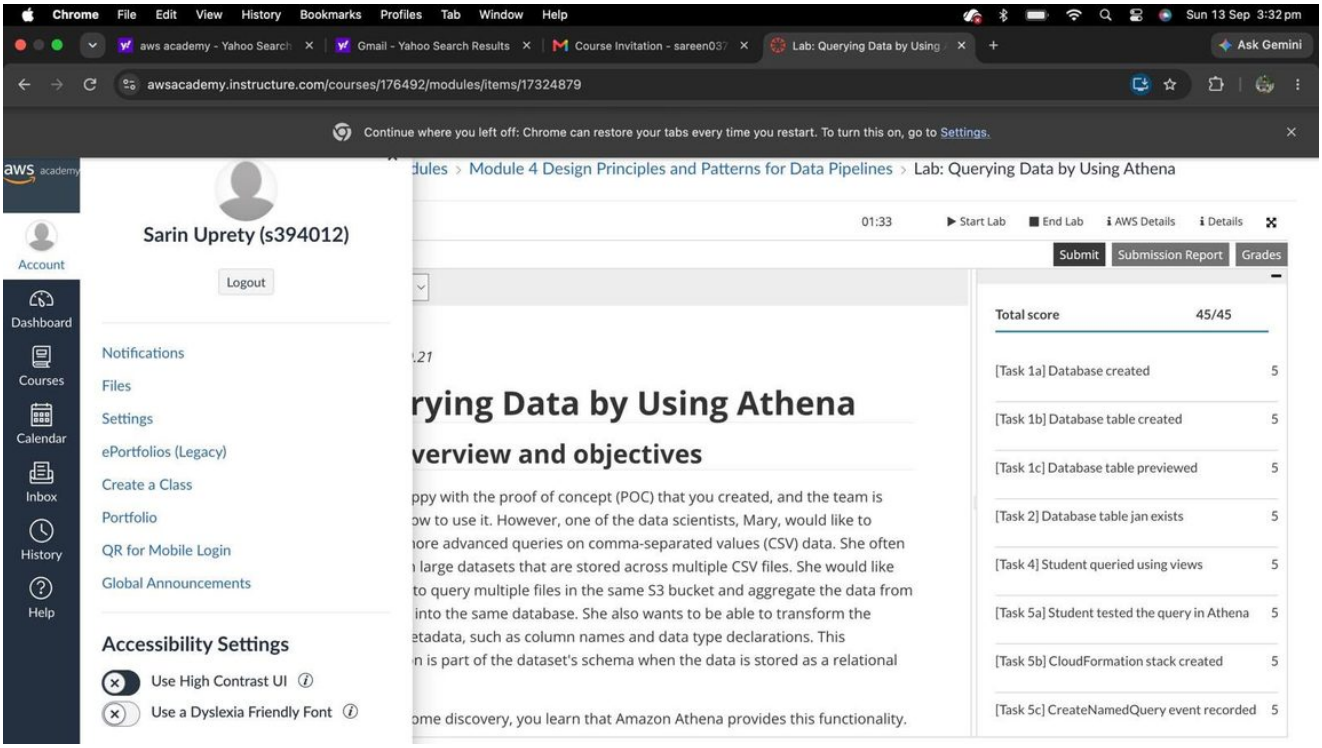
Converting the implicit gaps into explicit missing values was important because prior to this, if a reading was missing there was no row for that time, which made it difficult to determine how much data was missing. Once this full five-minute time grid was created, we were able to determine the 14,519 gaps and accurately measure the missing data. In Assessment 1 I was told there was no evidence of my work at all. I don't think I communicated / documented my work clearly. This time I have been more involved in the data collection and storage and my work is cited.

I want to be more consistent with my contribution and have evidence of what I am doing for Assessment 3. I will also make sure that I check the data properly before drawing any assumptions and speak to the group whenever there are issues/changes. I think this will make it easier for me to contribute and easier to see and check as I had faced the same problem with Assessment 1.

Sarin Uprety

Student ID S394012 · Data and Feature Engineer · Epics: Data Processing, Feature Engineering

Laboratory 1 Completion Evidence



Lab: Querying Data by Using Athena (Module 4). Completed, total score 45/45. Sarin Uprety (s394012). Captured 13 September 2026, 3:32 pm.

Individual Contribution

Contribution	Evidence
Completed 5 stage cleaning pipeline at 5-minute resolution and created before and after measured table	notebooks/03_clean_mastermeter1.ipynb, Outputs/cleaning_report_mastermeter1.md
Diagnosed and repaired the cumulative energy register: a six-digit counter with six wraps corrected and validated to 99.7% with the energy integrated separately by Active Power. Closes Assessment 1 risk T3	notebooks/03, section 3 output; SCRUM-34
Established that the negative overnight register steps are real net consumption, not corruption and prevented 43% of the record from being mislabeled	Section 5 of the technical report
Constructed the 30-feature hourly store, rolling and lag all features back to avoid temporal leakage	notebooks/04_features_and_eda.ipynb, Datasets/processed/features_hourly.parquet
Published the feature dictionary and the wind speed exclusion at 52.77% missing, as expected in Assessment 1 risk T4	Supporting_Documents/feature-dictionary.md; SCRUM-19

Documented schema drift and year-boundary duplicate finding, design change	Section 4; SCRUM-15, SCRUM-16

Repository: github.com/Uttam1618/PRT661-solar-forecasting · commit history · pull requests · notebooks
Jira board: SCRUM board 1 · **issues:** SCRUM-33, SCRUM-34, SCRUM-35, SCRUM-19, SCRUM-16, SCRUM-15

Professional Reflection

The way that the register functioned was different to what was assumed in Assessment 1. We thought it was just differenced, but it is a six-digit cumulative register, which resets back to zero at 1,000,000. This taught me to never assume and always verify any assumptions based on the actual data if it needs to be used in cleaning or modelling. Six wraps were identified and corrected and checked against Active Power with 99.7% agreement.

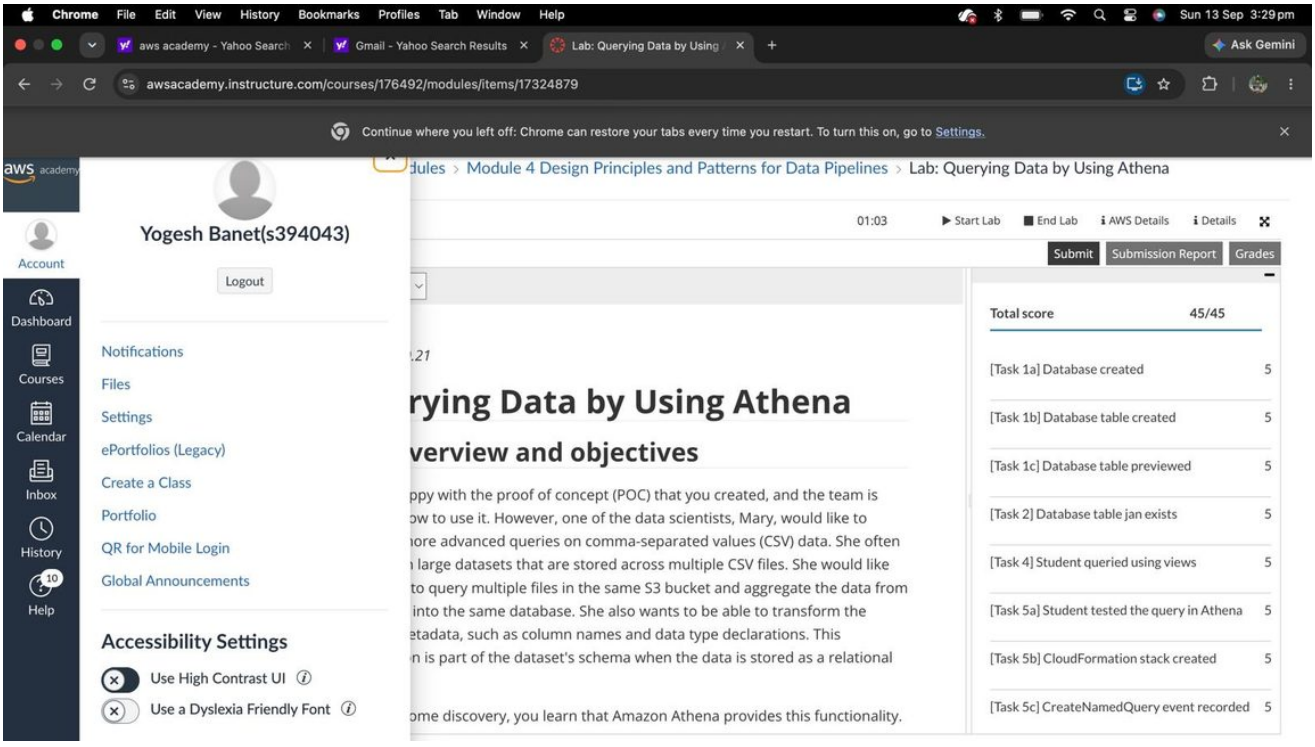
The order of cleaning is important because if data is aggregated before cleaning, the hourly averages could be impacted depending on what the incorrect values are. The error codes were added to the hourly data in the early process, and led to the unrealistic air temperature of -39.988 degrees C. I found out that raw data has to be “checked, cleaned” with realistic instrument and climate range limits, and then aggregated. I also learned not to interpolate the variable to be compared with the model because it may give a false agreement between the model and actual data.

There was not yet strong evidence of what I did in Assessment 1 as I didn't record my contribution clearly. This time, I can present some of the work that I did, such as correcting the register wraps, verifying the corrected outcome, finding the incorrect sequence in the cleaning, finding missing wind speed data, and determining what features to use. Assessment 3: Data assumptions will be validated earlier, evidence will be kept, value will not be filled in for the target, and data will be cleaned before aggregated.

Yogesh Basnet

Student ID S394043 · Modelling and Visualisation Lead · Epics: Analytics and Machine Learning, Visualisation

Laboratory 1 Completion Evidence



Lab: Querying Data by Using Athena (Module 4). Completed, total score 45/45. Yogesh Basnet (s394043), account name Yogesh Banet. Captured 13 September 2026, 3:29 pm.

Individual Contribution

Contribution	Evidence
Modelled and walk forward validated gradient boosting models at both horizons within four progressively expanding folds: 1 h MAE 3.61 kW, R2 0.980; 24 h MAE 6.96 kW, R2 0.913	notebooks/05_modelling.ipynb, Outputs/model_results.md; SCRUM-37
Three baselines implemented (persistence, seasonal naive, climatology) and determined that the seasonal naive is the best baseline at hourly resolution	notebooks/05, section 3 output; SCRUM-20
Recognized that training loss needs to be equal to the	Section 8 of the technical report; new risk T10

metric being reported; corrected T10 from -0.219 to +0.101 for 24 hours using absolute error loss (instead of squared error loss)	
Reported 24-hour interval, which is 'over-confident' at 0.706; reported quantile regressions for 80% prediction intervals by daylight and by night separately, and stated that the benchmark parity conditions	Outputs/model_results.md; Section 8; SCRUM-38, SCRUM-21, SCRUM-39
Developed 10 figures (with corrected autocorrelation on a true hourly grid), and created the error profile by hour of day and the dashboard concept	notebooks/04b, 05; Outputs/figures/fig1 to fig10; SCRUM-36, SCRUM-22

Repository: github.com/Uttam1618/PRT661-solar-forecasting · commit history · pull requests · notebooks
Jira board: SCRUM board 1 · **issues:** SCRUM-37, SCRUM-38, SCRUM-39, SCRUM-36, SCRUM-20, SCRUM-21, SCRUM-22

Professional Reflection

The 24-hour model was only marginally better when compared to the seasonal naive model. This gave me insight that the bulk of the prediction is not just from the model but also from the daily pattern in the data. The 1 hr model was more advantageous as it could be used to detect short term changes which the baseline model could not. I was told that it is not only important to have a good R2, but it is also important to have a good improvement over a proper baseline.

I reported outcomes that did not improve the project's appearance since the results must be truthful. I reported results that did not make the project look better. Our R2 was better than the benchmark study, but they had irradiance at the time of prediction, while we did not incorporate any future weather information to our model. So, it would be unfair to claim our model was more successful. Additionally, the 24-hour prediction interval only captured 70.6% of the time, rather than the expected 80%, and thus failed to meet the quality target.

In Assessment 1, I found it hard to prove what my contribution was as it was not clearly indicated. This time, I illustrated my efforts with the verification of the loss function, the problem with the mismatch of the metric, the investigation of the unusual fold, the verification of the autocorrelation analysis and the discussion of the limitations of the baseline and prediction intervals. For assessment 3, I will choose the evaluation method earlier, have a solid baseline, and make sure the model loss function is aligned with the evaluation metric, as well as carefully investigate my results when they don't make sense, pay more attention to daytime forecasting, and maintain evidence throughout the project.

Section 2. Final Project Update Summary:

Charles Darwin University, Danala (Darwin City) Campus
Group 4
PRT661 Data Science Practice
Semester 2, 2026
GitHub repository · Jira project

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1. Changes Since Assessment 2

Area	Previous State (A2)	Current State (A4)	Justification
Architecture	Notebooks 01-05: Forecast only	Notebooks 01-08 and dashboard, nightly battery plan and health checks (Fig. 1)	There was a stated purpose of battery decision.
Intervals	24 h coverage 0.706; gate failed	Conformal calibration: 80.1% / 80.0% and gate passes.	The interval needs to be reliable
Decision layer	Written guidance only	Battery model and 22:00 rule; saving 22.3% (A3)	Converts accuracy into cost.
Analytics, ML	Demand: 7 day mean, 54.5 kW error	Five models compared; boosting 24.9 kW; 39% - 70% of perfect.	Due to demand, the plan was limited.
Evaluation	Accuracy per fold	Cost against rules and the actual battery; bootstrapping intervals	Shows value (not just accuracy)
Monitoring	None	Daily checks, drift check, monthly refit	Detects degradation
Deployment	Dashboard deferred	Streamlit, 6 pages, nightly schedule (Fig. 13)	Puts the plan before the operator
Governance	1 of 8 gates failing	All gates pass, risk review, AI disclosure	Continuous evidence trail

2. Team Contribution Summary

Team Member	Major Contributions Since Assessment 2
Uttam Shrestha (s396560) Project Lead	Calibration, battery model, dispatch (PR #2); monitoring (PR #6); dashboard (PR #7); planning, report
Abhisek Tamang (S396739) Data Engineer	A3 architecture, README (PR #4); diagrams 09, 10, risk review, AI disclosure (PR #10)
Sarin Uprety (S394012) Data and Feature Engineer	A3 workflow diagram (PR #3); run_pipeline.py and final README (PR #11)
Yogesh Basnet (S394043) Modelling and Visualisation Lead	A3 video and deck (PR #5); demand models, notebook 07 (PR #8); rubric review

Evidence Locations

Evidence	Repository Location
Notebooks 06, 07, 08	notebooks/
Dashboard and screenshots	dashboard/, Outputs/figures/dashboard/
Results, figures 11 to 18	Outputs/
Architecture, governance	Architecture_Diagrams/, Supporting_Documents/
Planning and review	Jira SCRUM-44 to 73; pull requests #2 to #11

Section 3. Final Technical Development

1. Scope, and what changed since Assessment 2

Assessment 2 concluded with a working forecast: gradient boosting gave the Master Meter 1 output forecast with MAE of 3.66 kW one hour ahead and 6.96 kW 24 hours ahead with 80% intervals. It was also capped off with an honest gap. The forecast was the deliverable, the decision it existed to support, when the site battery should charge and discharge, was outside the system with the operator. 70.6% of daylight outcomes were achieved in the 24-hour period, one quality gate was not passed. The dashboard and planned operation were postponed.

All of that has been since then and the project has been moving across that boundary. Assessment 3 (notebook 06) calibrated the intervals, created a battery model based on the 2024 and 2025 site telemetry data and simulated a nightly charging plan against simpler rules. Assessment 4 enhanced the poorest aspect of that plan, the demand forecast (notebook 07); implemented the monitoring, drift detection and retraining necessary to bring a deployed model to life (notebook 08); and presented that plan to the operator via an interactive dashboard. This section records just that work. Table 1 details the resolution of the priorities set at the completion of Assessment 2.

Assessment 2 priority	Outcome	Where
Recalibrate the prediction intervals	Delivered. Daylight coverage at 1 h 80.1% and at 24 h 80.0%, spread between months halved.	Section 6.1
Add the dispatch layer to the 2024 telemetry	Delivered and enhanced. Saving 22.3% (A3), 22.4% with the A4 demand model; value of the forecast $3,419 to $3,943.	Sections 6.2, 6.3
Interactive dashboard	Delivered. Six pages (battery rule is rerun live for any setting).	Section 7
Update extract and automate the pipeline.	Partly. A nightly schedule is defined, failure handling and a cloud mapping are defined, the portal's Master meter 1 extract is still on 23 August 2025, so all evaluation stops there.	Section 10
Numerical 24 h weather prediction	Not delivered. DKASC does not provide forecast weather and a licensed archive of weather forecasts for Alice Springs was not possible. It was also determined that demand was greater than effort was, and thus effort moved to the A3 decomposition (Section 6.2).	Section 12
The temperature derating is separated from the angle of incidence derating.	Not delivered, for the same reason, it would "refine" the solar model, no longer the limiting factor in the decision.	Section 12

Table 1. Assessment 2 priorities and how each was addressed. Two were intentionally deprioritised not discarded without notice on evidence.

2. Final system architecture

The system is constructed as shown in Figure 1. The eight-layer structure suggested for Assessment 1 and validated for Assessment 2 remains the same, with the additions below the solar forecasting layer colored by the assessment that produced it. The project deliverable is no longer a forecast, but three products created together each evening: the solar forecast with a calibrated interval, a demand forecast for the planning hours of the next day, and the amount of off-peak grid energy the battery should store tonight, along with health checks, which say whether or not those numbers can be relied on.

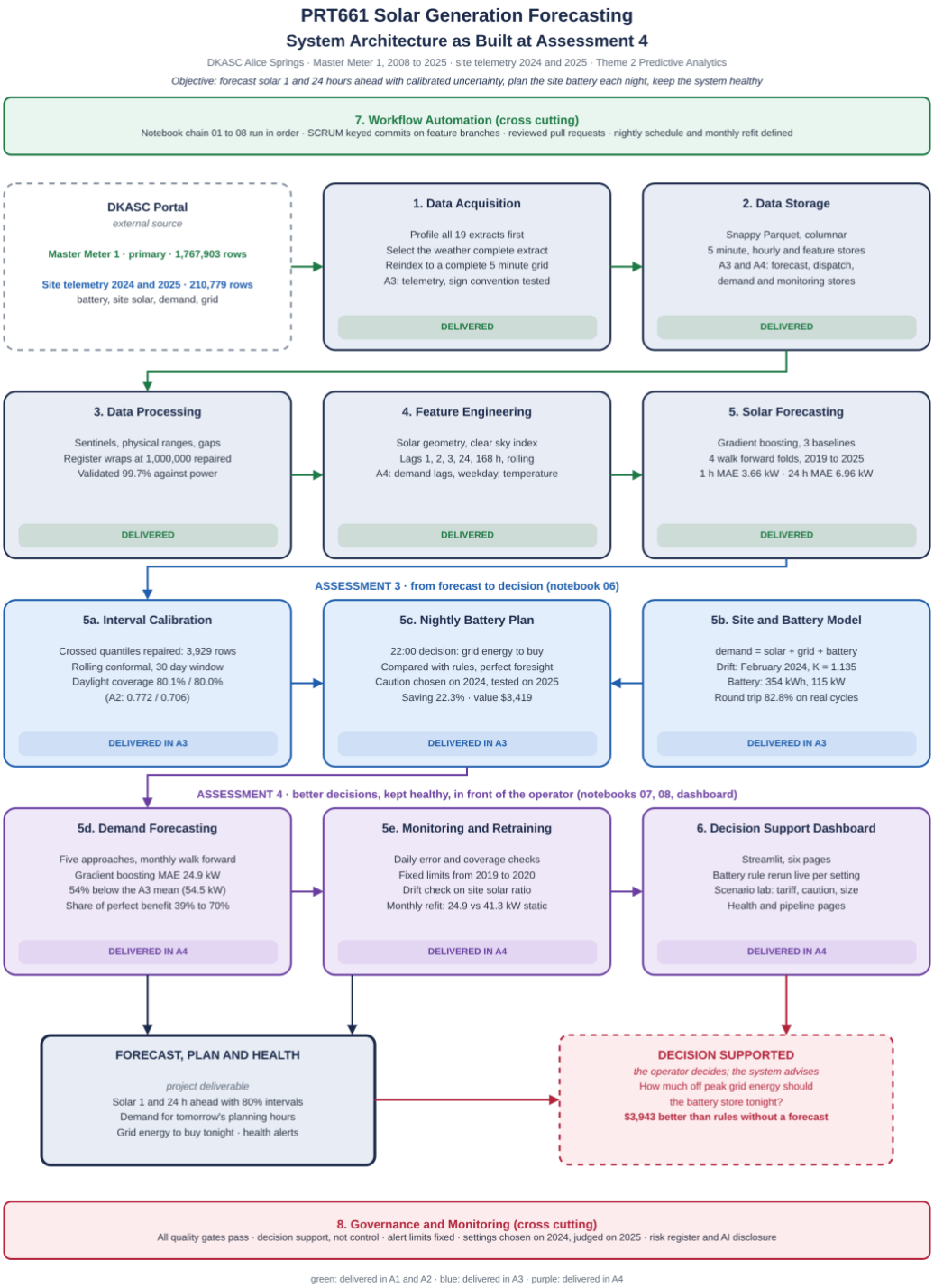


Figure 1. System architecture as built in Assessment 4. Green layers were delivered in A1, A2, Blue in A3 (notebook 06) and Purple in A4 (notebooks 07 and 08 and the dashboard). The choice is up to the operator: the system gives an indication of what storage of grid energy is recommended and how certain it is.

Three design decisions from Assessment 2 were reconsidered and retained. Not a database server, but parquet with DuckDB. The biggest new store, the calibrated forecasts, is 5.4MB and the hourly simulation table is 12,216 rows - a server would only add operating cost, but nothing else. Notebooks are implemented as the unit of work and files as contracts. The notebook reads only the stores written by the notebook in front of it and any stage can be rerun and checked in isolation (Table 2). Decision

support, NOT control. The system does not send any command to the battery; Section 11 discusses the importance of this boundary for governance.

3. Complete implementation

All work since Assessment 2 is committed and merged to the main branch via reviewed pull requests: notebook 06 for calibration, notebooks 01-08 for the battery model and the nightly plan (PR #2), notebook 07 for demand forecasting (PR #8), notebook 08 for monitoring, drift detection and retraining (PR #6), the dashboard and its battery module (PR #7), and run_pipeline.py for running notebooks 01-08 in order and checking the 39 expected outputs (PR #11). The contract between the stages is given in Table 2: what each stage reads and what it writes. Since each stage only looks at files created by the previous stages, each stage can be re-examined individually, and a check of the current repository succeeds.

Stage	Notebook or module	Reads	Writes
Forecast export (A3)	05_modelling	features_hourly	forecasts_hourly: 113,677 out of sample forecasts, 2 horizons, 4 folds
Calibration, battery, dispatch (A3)	06_bess_dispatch	forecasts_hourly, raw telemetry at site for 2024 and 2025	forecasts_calibrated, dispatch_hourly, dispatch_results.md and figures 11-14.
Demand models (A4)	07_demand_forecast	dispatch_hourly, mastermeter1_hourly_clean (temperature)	demand_forecasts, figures 15 and 16, demand_model_results.md
Monitoring and retraining (A4)	08_monitoring_retraining	forecasts_calibrated, demand_forecasts, raw site telemetry	monitoring_daily, monitoring_alerts.csv, monitoring_results.md, figures 17 and 18
Decision support (A4)	dashboard/app.py, battery.py	the processed stores and Outputs only	nothing: Read only, with CSV export.

Table 2. Data contracts between stages added since Assessment 2. Store live in Datasets/processed (not under git); results and figures in Outputs (committed as evidence).

4. Deployment

The system is deployed as a reproducible system that operates on one machine, not yet as a hosted service. With repository and the DKASC files, anyone can rebuild every store, table and figure using python run_pipeline.py, or check that they are there and up to date using python run_pipeline.py --check. Then the decision support dashboard begins by using pip install -r dashboard/requirements.txt and streamlit run dashboard/app.py. It can only read the processed stores, keeps no raw data and requires no credentials, and can rerun the nightly battery rule live for any setting the operator wishes (section 7).

There are two aspects of deployment that are not built – the nightly schedule that would run the chain at 21:00 each evening and a live data feed to replace the periodic DKASC download. Section 10

discusses the mapping of the same steps to scheduled cloud jobs (Figure 13), existing failure responses, and what remains to be done to allow the system to run unattended.

5. Data engineering since Assessment 2

Forecast export. Only fold summaries were retained for Assessment 2. Notebook 05 is now saving each out of sample forecast, along with the quantiles, 113,677 rows over both horizons and across all four walk forward folds (SCRUM-44). Each subsequent step is based on predictions the model makes without knowing the result, making dispatch evaluation honest.

Site telemetry. The five site level columns included in the yearly 2024 and 2025 extracts are those identified through profiling during Assessment 2, specifically: Battery power, Battery state of charge, Site Demand, Total site solar and Total site grid exchange. Notebook 06 uses 210,779 5-minute rows, none of which have duplicate timestamps, and 0.44% to 4.33% of which have missing values per column, and aggregates the data to 17,568 hours, discarding the 17,500 hours with fewer than 10 out of 12 readings. State of charge is assumed to be the level at the end of each hour, it is not a flow.

The sign convention was tested, NOT assumed. It is not specified in the extract that which way is positive for the battery and grid. The four possible conventions were tried out against the energy balance. Only demand = solar + grid + battery closes: the mean gap is 2.82 kW, and 96.3% of the hours balance within 2 kW, whereas the other three have gaps of 26.7 kW or more. The resulting average day is shown in figure 2; solar energy is in excess from approximately 10:00 to 14:00, with the battery charging and discharging in the evening.

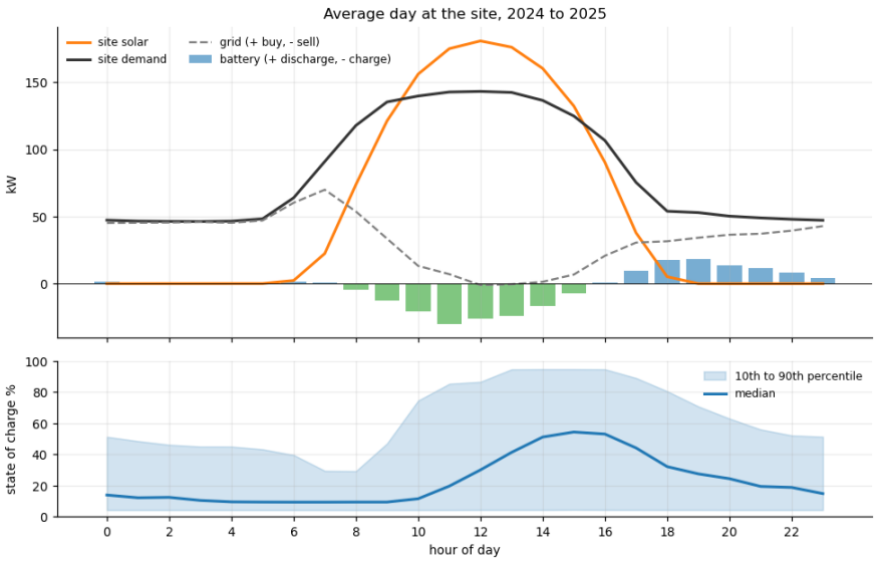


Figure 2. Average day at the site from the telemetry after the sign convention test (2024 to 2025). What a battery plan takes advantage of, is the midday surplus and the evening deficit.

Structural break identified and corrected. The ratio of whole site solar to Master Meter 1 was a median of 2.21 in January and February of 2024, and then declined to 1.13 in March, remaining in a range between 1.09 and 1.14 through August of 2025. This is a change in site configurations not weather. This scale factor, which transforms a Master Meter 1 forecast to a whole site forecast, was therefore fitted in March 2024 (K = 1.135), and tested from April 2024 onwards, with a correlation of 0.995 and a scaling error of 3.4 kW. As demonstrated in section 6.4, this break would have been automatically detected by the drift check introduced in Assessment 4.

The simulation table. Actual solar and demand, the scaled solar forecast and its calibrated low edge, the demand forecast and the tariff period are aligned on one hourly grid from 1 April 2024 to 22 August 2025: 12,216 hours. Short gaps of up to three hours are interpolated (62 hours in total) and the 80 hours with no solar forecast fall back to the same hour on the previous day. Every strategy is scored

on the same 12,154 hours that have a recorded grid reading, so the comparison with the battery as it was actually operated is like for like.

Assessment 4 additions. The demand features added to Notebook 07 are: demand at the same hour on day minus 1, 2, 7, 7 day mean, temperature at the same hour yesterday from Master Meter 1 weather station, hour, weekday, weekend and month. Notebook 08 creates a daily monitoring table for 2,383 days from January 2019, and then reprocesses the raw telemetry to calculate the site solar ratio used for drift detection.

6. Analytics and machine learning

Four analytical components have been added since Assessment 2. Interval calibration results in a solar forecast uncertainty that is trustworthy (6.1); the battery model and the nightly rule convert the forecast into a decision (6.2); five demand models are compared in order to make the forecast as strong as possible when it is used for the decision (6.3); monitoring, drift detection and retraining keep the models healthy once deployed (6.4).

6.1 Calibrating the prediction intervals

Two issues were addressed prior to the use of the interval for planning. First, the 10th and 90th percentile models are trained independently and sometimes the 10th percentile was greater than the 90th percentile: 3,929 rows (3.5%) experienced the lower quantile being greater than the upper quantile, and this happened most at night when both are close to zero. The lower edge was set at the smaller value and they were repaired. Second, the coverage bounced around between folds from 0.744 to 0.846, while the last 24 hours fold was 0.707.

Fix is rolling conformal calibration (Angelopoulos & Bates, 2023). A nonconformity score is a record of the distance between the actual and the interval (negative if the actual is within the interval). The scores for each target day D from D-31 to D-2 days, all known before the forecast for D is made, provide a correction Q at the finite sample level of ceil((n+1)(1 minus alpha))/n) where alpha = 0.20. The interval turns into [low-Q, high+Q]. A positive Q will increase an overconfident interval, and a negative Q will decrease a cautious interval. The window rolls due to seasonal and regime changes as the recommended adaptive form in the context of distribution shift (Gibbs & Candès, 2021). Daylight hours are calibrated only as daylight coverage is the Operationally meaningful measure set in Assessment 2.

Horizon, fold	Coverage before	Coverage after	Width before (kW)	Width after (kW)
1 h, fold 1 (2019 to 2020)	0.744	0.797	19.9	20.6
1 h, fold 4 (2024 to 2025)	0.780	0.799	24.5	25.0
24 h, fold 1 (2019 to 2020)	0.780	0.796	38.2	38.8
24 h, fold 3 (2023)	0.846	0.810	60.1	58.6
24 h, fold 4 (2024 to 2025)	0.707	0.793	45.9	48.4
All folds, mean distance from 0.80	0.038	0.005		

Table 3. Daylight coverage of the nominal 80% interval before and after calibration (notebook 06). Overall coverage is now 80.1% at 1 h and 80.0% at 24 h.

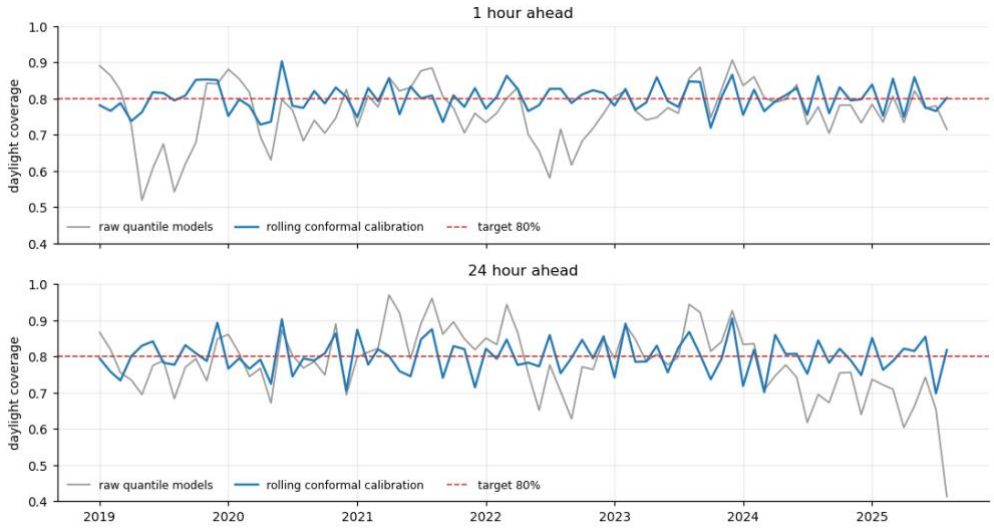


Figure 3. The monthly daylight coverage before rolling conformal calibration (grey) and after (blue). The month to month spread falls from 0.080 to 0.038 at 1 h and from 0.093 to 0.048 at 24 h.

Calibration interval width varies by less than 3kW in most folds, giving the operator a true interval with little loss in sharpness. A total of 305 rows had a strongly negative Q, and these were all collapsed to the median. Now, the A2 coverage gate (daylight coverage equals 0.75 to 0.85) passes at both horizons.

6.2 From forecast to decision: the battery model and nightly plan

A battery model fitted to the site.

The battery parameters were ascertained from the telemetry - not from a datasheet. Capacity is based on the energy required to charge by one percent (3.89 kWh over 2,411 charging hours) and the energy recovered from one percent discharge (3.22 kWh over 4,340 discharging hours); their geometric mean is 354 kWh, and their ratio is a round-trip efficiency of 82.8% on real cycles. The usable range is 2.2% to 94.8% state of charge. The 99th percentile charging power: 114.8 kW. The discharge percentile (63.6 kW) represents the minimum night demand the battery is required to meet, not the battery's ability, so the charging limit is applied both to the charging and discharging. The standby draw is 0.72 kW.

The decision rule.

An illustrative time of use tariff (22:00 to 06:59 at $0.15 per kWh, shoulder at $0.25, and peak time of day at $0.40, feed in $0.07) allows the battery to make two contributions to the bottom line: it stores solar energy off peak when it would otherwise be exported at 7 cents per kWh and it buys off peak electricity from the grid to power the evening peak. Both of them strive for the same space The question asked each evening at 22:00 in the plan is thus: how much grid energy to buy tonight. It predicts the solar excess over the demand for the next day from 07:00 to 15:59, converts this to the energy that this will provide to the battery and purchases only the remaining space off peak. Excess solar is stored during the day, and the battery is discharged in the peak.

The solar input uses a cautious forecast: the median moved towards the calibrated low edge by a caution factor c, so that solar plan = median minus c times (median minus low edge). This is where the uncertainty comes in the decision. In Assessment 2, it was posited that under preparing is expensive; c makes that a tunable, testable setting for an operator.

Strategy	Energy cost ($)	Saving vs no battery	Peak import (MWh)
No battery	128,709		146.2
Recorded battery, as operated	116,067	9.8%	106.0

Strategy	Energy cost ($)	Saving vs no battery	Peak import (MWh)
Solar only, no forecast	106,478	17.3%	71.0
Always fill overnight	103,418	19.6%	22.8
Forecast plan, caution 0.5	**99,999**	**22.3%**	
Perfect foresight	96,136	25.3%	22.8

Table 4. Strategies scored on the same 12,154 hours, 1 April 2024 to 22 August 2025 (notebook 06). The forecast plan captures 47% of the benefit perfect foresight adds and saves $3,419 more than the better rule without a forecast.

What limits the plan.

Replacing each forecast with the truth in turn located the weakness (Figure 4, left). With both forecasts the median plan captured 41% of the perfect foresight benefit. Perfect solar took it to only 52%, perfect demand took it to 85%. It was the demand forecast that was the constraining factor, not the solar forecast. This one outcome led to Assessment 4 priority (Section 6.3). The caution factor was selected based on the data from 2024 (c = 0.5) and evaluated on 2025 data, which was not used in the selection process: it captured 49% there as opposed to 48% for the median. The larger the battery the more the savings are, and at all battery sizes, the forecast remains positive (saving of 13.3%, 22.3% and 26.9% at half, one and two times the current battery size).

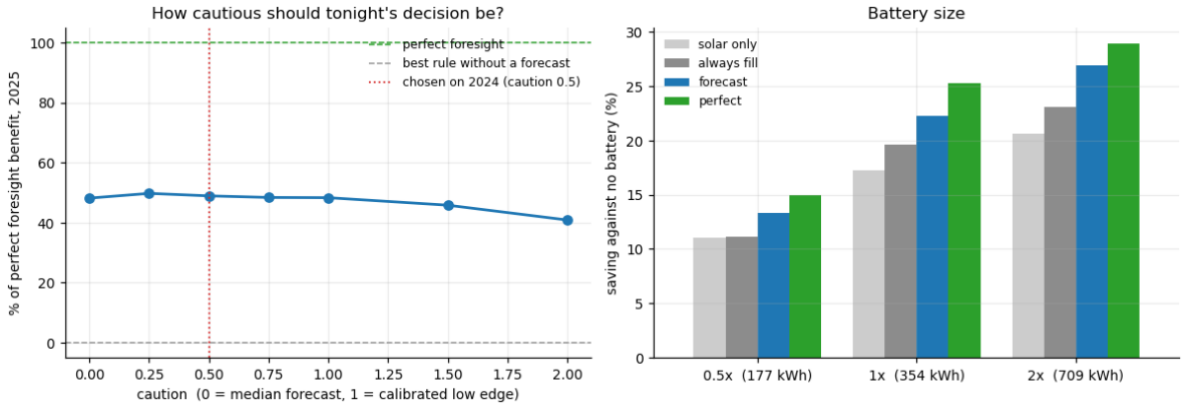


Figure 4. Left: fraction of PF benefit received in 2025 based on a 2024 choice of caution level. Right: saving against no battery, according to battery size (equal power/size ratio).

Figure 5 illustrates the plan on an ordinary week. It consumes less overnight before sunny days and charges the battery before cloudy days. It has two obvious errors which are instructive. The demand forecast, a 7 day mean, expected that on Sunday 25 May a working day was to be used and bought 186 kWh which a perfect foresight would not have bought. The forecast did not catch the clouds and was 124 kWh below the cloudiest day of the week, 2025 median 1293 kWh on Tuesday 27 May. The first error is erased by Assessment 4.

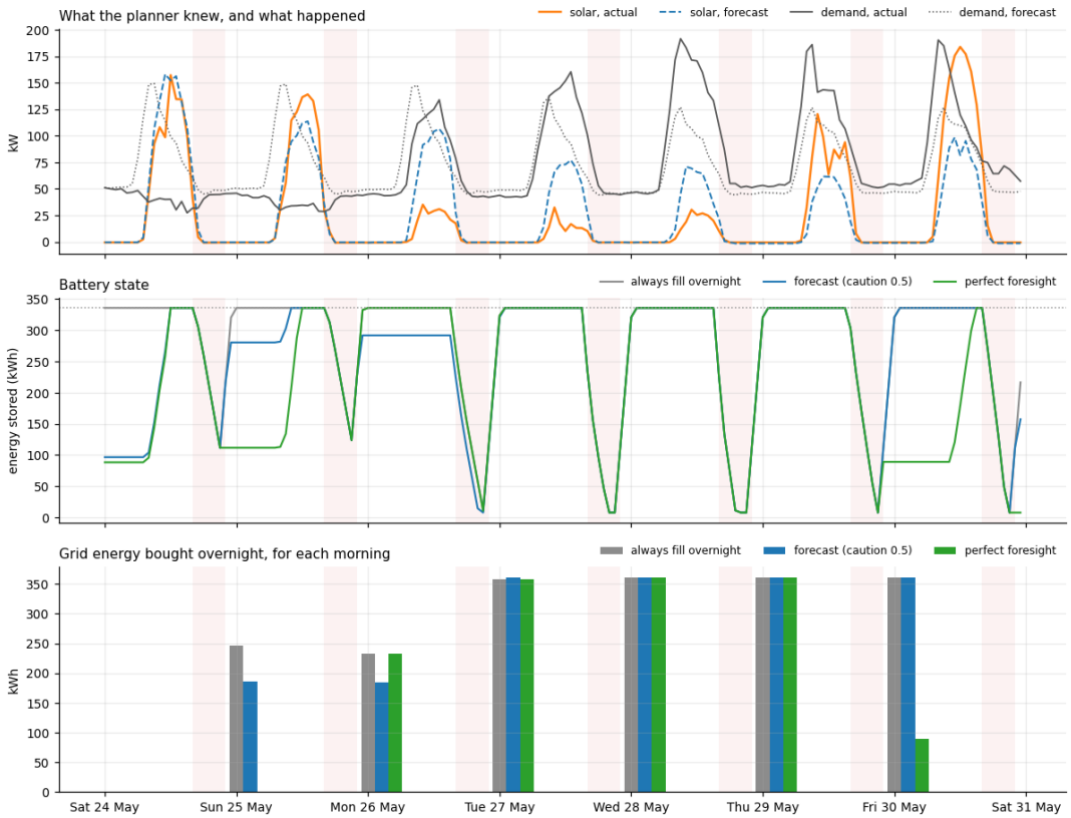


Figure 5. Forecast and actual solar and demand (top) and the energy stored under each strategy (bottom) for example week (24-30 May 2025).

6.3 Demand forecasting: five approaches compared

Notebook 07 compares five approaches to predicting tomorrow's demand for a planning window (07:00 - 15:59) based on only the information available the evening before (22:00). Two of them are naive references (same hour yesterday, same hour last week); one is the A3 seven day mean; two are learned models. One hot hour, weekday, month: is the interpretable choice: every effect is additive coefficient - Ridge regression. Histogram gradient boosting (Pedregosa et al., 2011; Ke et al., 2017) can learn interactions like the weekday shape changing over the hours and over the seasons and was the best solar model in Assessment 2. The same nine features (Section 5) are used for both the learned models.

The evaluation process is walk forward with monthly refits on an expanding window following Hyndman & Athanasopoulos (2021): forecast for each month using only data from previous months and then add that month of data to the training set. Fourteen monthly refits provide a total of 10,032 out-of-sample hourly forecasts up until 22 August 2025. As only these hours count towards the battery decision, the selection metric is the planning hour MAE, with all hour figures reported alongside.

Model	MAE (kW)	RMSE (kW)	Bias (kW)	MAE weekday	MAE weekend	Better than A3
A3: 7 day same hour mean	54.5	68.1	minus 0.4	42.3	85.5	
Same hour yesterday	47.5	73.6	minus 0.2	42.7	59.6	12.9%
Same hour last week	34.9	52.6	minus 0.1	42.2	16.2	36.0%
Ridge regression	30.3	43.2	minus 2.8	33.4	22.3	44.5%
Gradient boosting	**24.9**	**38.1**	minus 4.0	**28.4**	**16.2**	**54.3%**

Table 5 Notebook 07: Accuracy of the demand forecast, planning hours 07:00 to 15:59, out of sample, July 2024 to August 2025. All hours MAE: 26.5 kW (A3 mean) and 13.9 kW (gradient boosting)..

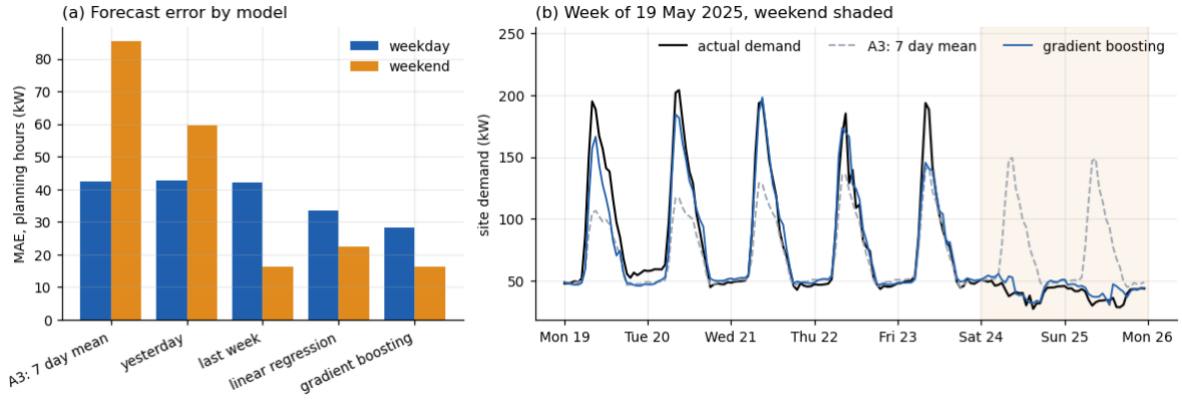


Figure 6. Weeks of planning error by model by day type (left) and one week forecast error for demand (right).

The table is self-explanatory once broken up by day type. The site is mostly closed at the weekend, with a seven day mean incorporating five working days in every Saturday and Sunday, giving the weekend error of 85.5kW - twice the weekday error. Same hour last week corrects for weekends (16.2kW) not weekdays. On both gradient boosting works best, as it does know which weekday it is, and still uses recent levels and temp. Its small negative bias indicates that it underforecasts by 4 kW on average, which is to the "safe side" in terms of cost as any shortfall is met at the higher price, but not lost.

It isn't about the accuracy, it is about the battery plan. The same solar forecast and caution is therefore applied to each demand forecast and scored over the same hours (Table 6).

Battery plan with demand from	Energy cost ($)	Saving vs no battery	Value vs best rule ($)	Share of perfect
A3: 7 day same hour mean	89,508	20.8%	2,209	38.9%
Same hour yesterday	89,592	20.8%	2,124	37.5%
Same hour last week	88,528	21.7%	3,188	56.2%
Ridge regression	88,381	21.8%	3,335	58.8%
Gradient boosting	**87,773**	**22.4%**	**3,943**	**69.5%**
Perfect demand (ceiling)	87,151	22.9%	4,566	80.5%

Table 6. Each demand forecast, battery plan rerun with solar caution 0.5, from July 2024 to August 2025. No battery costs $113,072; perfect foresight $86,044.

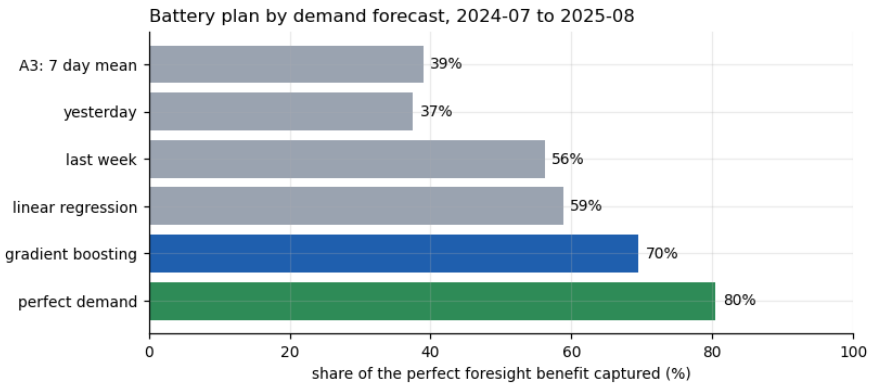


Figure 7. Foresight benefit share for the battery plan per demand forecast.

Gradient boosting increases the proportion of the benefit of perfect foresight from 39% to 70% and the value of the forecast from $2,209 to $3,943. This is 74% of the gap closed between this solar forecast and what a perfect demand forecast would provide (69.5 – 38.9) / (80.5 – 38.9). Note that accuracy is not directly proportional to value: same hour yesterday is 13% more accurate than the A3 mean, but a little less valuable because the errors tend to be bunched around Mondays (120.0 kW, forecast from a quiet Sunday) and Saturdays (105.4 kW, forecast from a working Friday), the nights that make the greatest difference to the overnight decision. This is why the project is measured in dollars, and not just kilowatts.

6.4 Monitoring, drift detection and retraining

An accurate model at the time of evaluation may degrade silently because of changes in weather regimes, equipment or site use (Gama et al., 2014; Sculley et al., 2015). Notebook 08 is a notebook with three safeguards and an evaluation of each on history.

Performance monitoring. The 24 hour daylight solar forecasts are scored daily with respect to a 30-day rolling window on a two metrics. The normalised MAE (MAE/mean daylight output) triggers an error alert if the normalised MAE is higher than 0.354, the 99th percentile for 2019 and 2020. The calibrated interval is alerted if the coverage is greater than 0.70 and 0.90. Any downward trend can be masked by a shifting baseline, and these limits were never reviewed, but held constant for the first two years.

Year	Days	Normalised MAE	Coverage	Error alert days	Coverage alert days
2019	345	0.087	0.801	0	3
2020	366	0.135	0.795	8	15
2021	342	0.157	0.803	3	18
2022	365	0.176	0.800	0	13
2023	365	0.174	0.808	24	7
2024	366	0.179	0.794	8	5
2025 (to 22 Aug)	234	0.138	0.797	0	18

Table 7. 24-hour solar forecast daily monitoring 2,383 days (notebook 08). The calibrated coverage falls in the range of 0.794-0.808 for all the year.

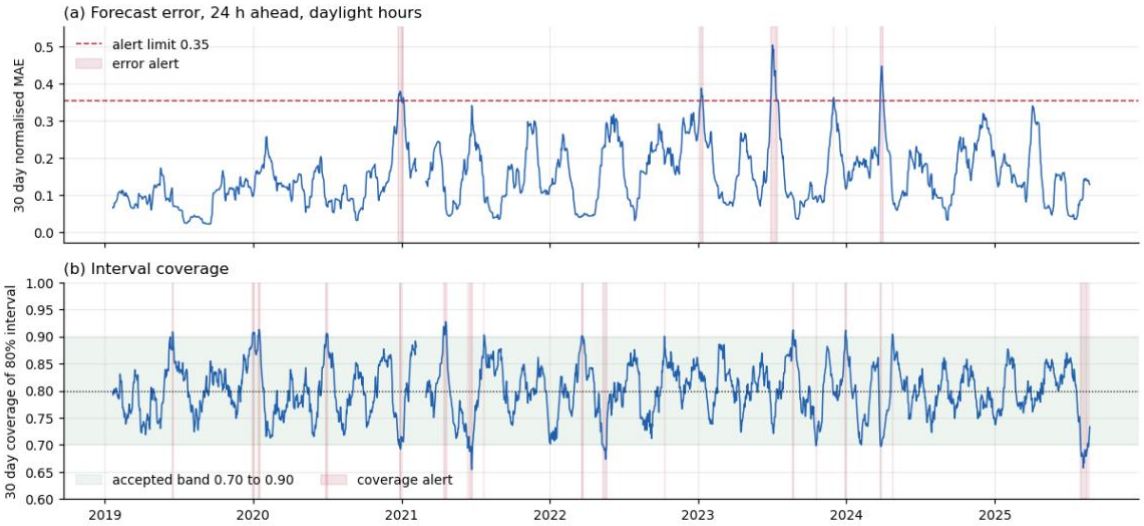


Figure 8 The 30-day normalised error (top) and interval coverage (bottom) are plotted along with the alert limits and the 28 alert episode.

The checks resulted in 28 alert episodes in 6.5 years, or approximately 4 episodes per year – a manageable number for a human review team to handle. The longest coverage was 17 days from 31 July to 16 August 2025 as the 30-day coverage fell to 0.658 on 6 August – too narrow to sustain a run of unusually variable days – and recovered to 0.733 on 22 August when the rolling calibration widened the interval while no manual action was taken. The error alerts are concentrated in two seasons, July (14 of the 2023 error days), when average output is at the lowest, which leads to a relatively high normalised error when a few cloudy days occur, and December to March (the convective cloud season), which Assessment 2 rated as the most difficult to predict.

Data drift detection. Model error is delayed. The input data may indicate this change earlier and ratio of whole site solar to Master meter 1 is also monitored. A drift flag is set once the 7-day median is more than 20% different from the reference (which is the median of the first 28 days of 2024: 2.24). The configuration change that Section 5 found by inspection, was applied to 2024, and the check fires on 28 February (seven day median 1.13, 49% below the reference). Once the reference is set to the new value (1.14) there is no more day marked as August 2025. The scale factor K was fitted on the exact same post change month and the check in hindsight confirms a decision that was made by hand in Assessment 3.

Retraining strategy. Three strategies were retested for the gradient boosting demand model for the period July 2024 to August 2025: static model trained once; refit monthly; refit when trailing error is 15% higher than normal.

Strategy	Planning hour MAE (kW)	Worst month MAE (kW)	Model fits
Static, trained once	41.3	95.6	1
Scheduled monthly refit	**24.9**	**53.9**	14
Triggered (error 15% above baseline)	29.1	59.9	3 (refits November 2024, January 2025)

Table 8. Retraining strategies for the demand model (notebook 08).

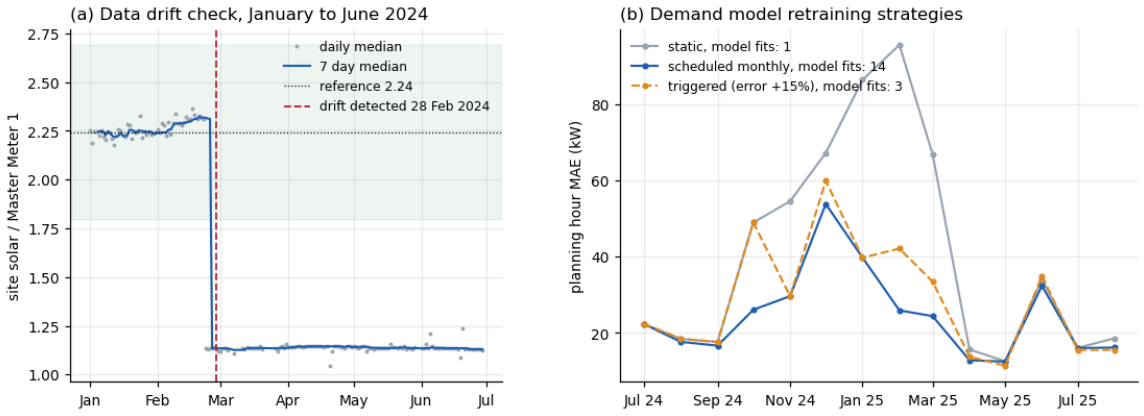


Figure 9. The drift check indicates the site change that occurred in February 2024. Right: Percentage of monthly planning hour error with each retraining strategy.

A static model goes to the dogs: its worst month error is 95.6 kW, almost twice the monthly refit. With triggered retraining, there are three fits but a month's delay for each one. One refit takes seconds on this data size, so there is no cost argument as to why or why not to trigger it, only that monthly refits are the operating choice with the triggered option being an early refit when an alert is present. No retraining of the solar model was performed during this time interval and monitoring above indicates that it did not need to be.

7. Decision support dashboard

The dashboard (Streamlit, Plotly) is designed for the site operator / energy manager, not for a data scientist. It does not store any raw data, doesn't require any credentials and contains only the processed stores and Outputs with the starting command streamlit run dashboard/app.py. It doesn't simply show saved results: that's its core principle. The battery rule from notebook 06 was implemented in a small module (dashboard/battery.py) and rerun live when an operator changes a setting. With the default settings, this module reproduces notebook 07 to the cent ($87,773.1) and notebook 06 within $12, which is just due to rounding of the battery parameters. Cache: A full scenario (5 strategies / 12216 hours) takes 0.10 to 0.16 seconds.

Page	Question it answers for the operator	Main interactions
Overview	Is the plan worth running and what is the savings gained?	Headline tiles with change against demand forecast (A3); cumulative and monthly saving by strategy
Battery plan by day	Was battery plan correct last night?	Clickable calendar of daily savings; decision indicated as space – estimated solar – grid energy to buy (CSV export); solar range (space) against actual (solar)
Scenario lab	What happens if the tariff/caution/battery is different?	Tariff and period sliders; tabs comparing caution levels, the five demand forecasts and battery sizes
Forecast models	Which demand model and where does it break down?	Errors per hour and month; switch between weekday and weekend; week explorer
System health	Can tonight's numbers be trusted?	Status Banner, coverage and error chart with limits and alert, retraining and drift summary
Data and pipeline	From where does each number come and is it new?	Pipeline diagram, inventory of all inputs, with size and date, and nightly run schedule

Table 9. Dashboard pages. Settings in the sidebar (demand forecast, solar caution, battery size) are per page, and default to values selected in the notebooks.



Figure 10. Overview page. The plan saves 22.4% when compared to the case with no battery, $3,943 when compared to the best rule without forecast and 70% of the perfect foresight benefit when compared to the A4 demand model.

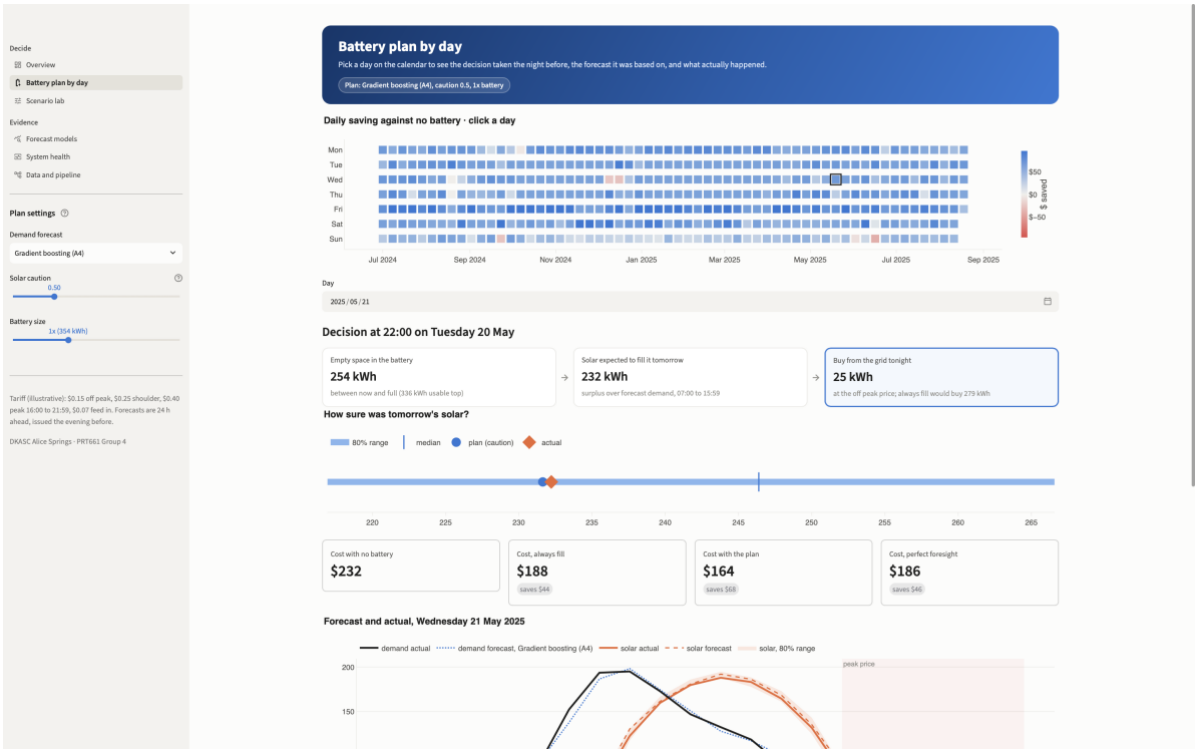
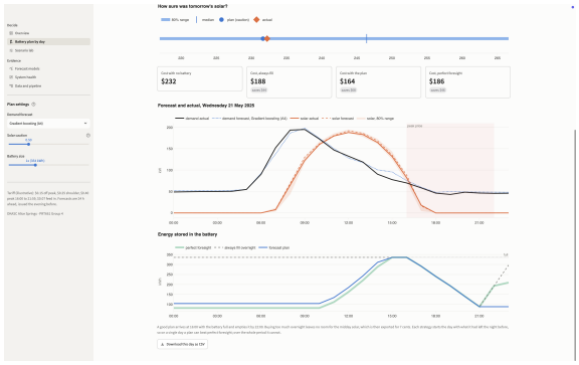


Figure 11. Battery plan on Wednesday 21 May 2025. The battery was at 22:00 the night before and had 254 kWh of space, with 232 kWh worth of solar expected to fill it, so the plan purchased 25 kWh, but with a 279-kWh purchase with always filling.

The other pages are displayed in Figure 12. The bottom half of the daily page is a comparison of the forecast with what occurred, and includes the energy stored under each strategy, the Scenario lab to

rerun the plan for any caution level, demand forecast, battery size or tariff, the System health page with status for tonight and monitoring charts with their limits, the retraining comparison, the drift summary and a filterable list of alert episodes.



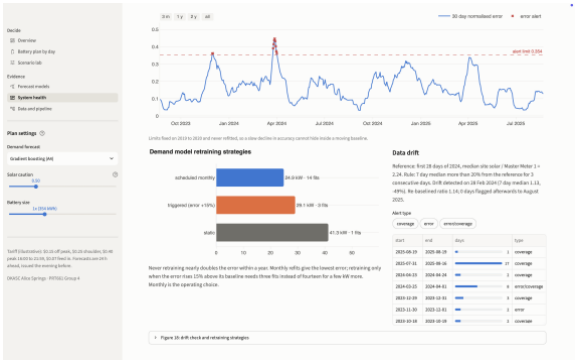
(a) Battery plan by day: forecast vs actual and energy stored - CSV export



(b) Scenario lab: Share of the perfect foresight benefit of solar caution



(c) System health: banner, latest checks, interval coverage & alerts



(d) System health: error with alert limit, retraining strategies, drift and alert episodes

Figure 12. Further dashboard pages. All six screenshots are saved to Outputs/figures/dashboard.

The rules of visual design are as follows: Each page has one meaning per colour; all charts have legends, units are labelled and hover values are given; Numbers are always expressed in plain language next to a chart; Status is expressed with words and an icon, never with colour only. It is not presented as a number, but as a three-step calculation, because the operators are more likely to trust and correctly override a number for a recommendation that has been shown to them.

8. System evaluation and performance analysis

Table 10 looks back at the quality gates established in Assessment 2 and introduces the quality gates required by the decision layer. All gates now pass; A2 - no more coverage failure, just a fix.

Gate	Threshold	A2	A4
Forecast performance	1 h MAE at most 5.0 kW; 24 h at most 9.0 kW	3.66 / 6.96 PASS	unchanged PASS
Interval coverage	Daylight coverage 0.75 to 0.85	0.772 / 0.706 FAIL at 24 h	0.801 / 0.800 PASS
Fold stability	MAE standard deviation with respect to mean is within 20%	10.8% / 11.5% PASS	unchanged PASS
Reproducibility	All numbers and figures regenerated from notebooks	10 of 10 figures	18 of 18 figures (checked in the dashboard with notebooks)

Gate	Threshold	A2	A4
Dashboard refresh	Updated within 60 minutes of new data	deferred	dashboard refresh: plan at 22:00, met by design: reads stores on start.
Decision value (new)	Plan beats the best rule without a forecast in a year not used to tune it		2025: 49% of perfect benefit PASS
Monitoring (new)	Daily checks, with fixed limits and a recorded response		28 episodes, response defined PASS

Table 10. Quality gates at Assessment 2 and Assessment 4.

Robustness. Three tests test the result to see if it is due to lucky guesses. Caution factor is tuned on 2024 and kept constant on 2025 (Section 6.2). The plus of the forecast lasts half or a double battery. The tariff was then manipulated with the A4 plan on the dashboard's Scenario lab from July 2024 till August 2025 (Table 11).

Tariff scenario	Saving vs no battery	Value vs best rule	Share of perfect
Default: $0.15 / $0.25 / $0.40, feed in $0.07	22.4%	$3,943	69.5%
Higher peak price, $0.55	30.6%	$3,500	61.7%
No feed in payment	23.9%	$7,012	72.5%
Cheaper off peak, $0.10	27.9%	$899	42.1%
Flat $0.25 all day	2.8%	minus $4,976	not meaningful

Table 11. Sensitivity of the A4 battery plan to the tariff (dashboard Scenario lab, gradient boosting demand, caution 0.5).

The pattern is educative. If the solar produces nothing one day, then it knows that the surplus that it can produce the next day is worth nearly twice as much, because it would have to pay more to use the space with grid energy. If energy costs are very low at off-peak times, it is very close to optimum to fill every night, and the forecast doesn't help much. In a flat tariff, the rule is incorrect: in a flat tariff, purchasing grid energy overnight only results in a 17% round trip loss, and the plan is worse than solar storage alone. The rule is valid for time of use tariffs only, and the tariff must be checked before deployment (Section 12).

Uncertainty of the headline results. One cost number for 14 months does not reveal the extent to which it might change due to different weather and demand patterns. So, each headline was resampled 5000 times using a moving block bootstrap (Künsch, 1989) which retains the weekly patterns and day to day dependence inherent in the data. The 95% intervals are given in Table 12.

Result (July 2024 to August 2025)	Estimate	95% interval
Saving against no battery, A4 plan	22.4%	20.4% to 24.8%
Value of the forecast over the best rule without one	$3,943	$3,190 to $4,700
Share of the perfect foresight benefit captured	70%	59% to 79%
Extra value from the A4 demand model over the A3 mean	$1,734	$1,160 to $2,300
Reduction in planning hour demand MAE, A4 against A3	29.6 kW	25.3 to 33.6 kW

Table 12. Moving block bootstrap intervals (7-day blocks, 5,000 resamples). Intervals are based on a re-run of notebook 07, the estimates are those of committed outputs.

None of the intervals contain 0, so the improvements are not simply due to a lucky interval: the plan is better than the best rule without a forecast, and the A4 demand model is better than the A3 mean, with high confidence. As is the nature of a forecast-based decision, the picture is mixed day by day, but the cost of the plan is less than 57% of days and more than 18% of days, and profits on the many common days outweigh losses on the few days where the forecast is wrong.

Scalability and performance. Data volumes are kept low: 1.4 MB per hour table and less than 15 MB for all of the processed data store. The number of hours is linear, and after a decade of such data they would still run in a second or so, and you'd add another site with just another table and not a new architecture. Refitting models takes seconds using a laptop. It's not compute, it's data access, for a live meter feed and a weather forecast, both outside the DKASC download portal.

9. Discussion of results and proof of impact

The easiest way to see first-hand the impact is to compare with how the battery was actually used. The battery cut produced a recorded reduction of 9.8% of the energy costs, with the forecast plan producing a cut of 22.3% of the energy costs over the A3 period (1st April 2024 to 22nd August 2025). The plan is $16,068 cheaper than the battery as operated on the same hours and tariff, or approximately $11,600 a year. The A4 plan costs $87,773 for the A4 evaluation period (July 2024 onwards) while the recorded operation costs $103,695, a difference of $15,922, or 154 days. The recorded battery discharged 66.5 MWh over the A3 period compared to 111-126 MWh under the strategies that charge from the grid overnight: the asset was under used, and a nightly plan is a low cost way to use it.

These figures have two worthy caveats. The tariff is a representative tariff and not the contract to be signed at the site, therefore the dollar amounts indicate the magnitude of the opportunity, not a definite saving (Table 11 illustrates how they fluctuate). The plan is simulated, that is, it is assumed that the battery will operate in exactly the same way as the plan indicates, whereas the actual operation that was recorded shows real constraints, including uses for the battery that this analysis does not observe, such as uses for backup.

Assumptions. The evaluation is based on five assumptions, some of which are bounded where possible. The battery is to be as planned exactly within limits of the measured power, capacity and efficiency. Constant round trip efficiency (82.8%) and no consideration of wear – (2) (3) The tariff is of the 3period time of use type; see Table 11, the conclusions apply to plausible prices, but do not to a flat rate. The drift check confirms that the scale factor K between Master Meter 1 and the whole site is stable as long as March 2024 and thus stable up to August 2025. The walk forward design requires that all forecasts be based on the same information that is available at 22:00.The 22:00 time is the information that each forecast's based on, which is enforced by the walk forward design. Assumptions 1 and 2 work to the benefit of the plan and make the simulated saving the most significant reason why the plan should be tested in a shadow mode trial before it is implemented.

Recommendations supported by the findings.

- Execute the nightly plan using gradient boosting demand forecast and solar caution 0.5, using the operator's approval for the overnight purchase. This is representative of 70% of the benefit that perfect knowledge of tomorrow would provide.
- Re-calibrate the demand model every month and continue to monitor daily and check on drift, taking action if an alert is showing for over a week.
- Ensure site is on a time of use tariff and change the illustrative prices before deployment; if flat tariff, then use solar only storage.
- The remaining 30% gap to perfect foresight is now to be addressed in a second step by investing in a weather forecast input to the solar model and an optimiser for the decision (Section 12) respectively.

Discussion. The project's central finding changed between assessments, and the method is what made the change visible. Assessment 2 reasonably expected that better solar forecasting, through numerical weather prediction, would be the largest improvement. Evaluating in dollars against perfect foresight showed instead that the demand forecast was the weak link, and that a model which simply knows the day of the week was worth more than any plausible solar improvement. A second lesson concerns uncertainty: calibration only affected the width of the interval by a few kilowatts, but it did make the interval trustworthy, and the caution factor took the trustworthy interval and made it measurable. It is the very concrete evidence: each of the recommendations above leads to a table in this section generated by a dedicated notebook.

10. Deployment readiness

Figure 13 shows how the system would run in operation. Five scheduled jobs are all mapped to the same notebook chain, one job writing each zone which is read by the next job. The schedule is fixed: ingest at 21:00, forecast at 21:15, health checks at 21:30, plan at 22:00 and a demand model refit on the first of each month.

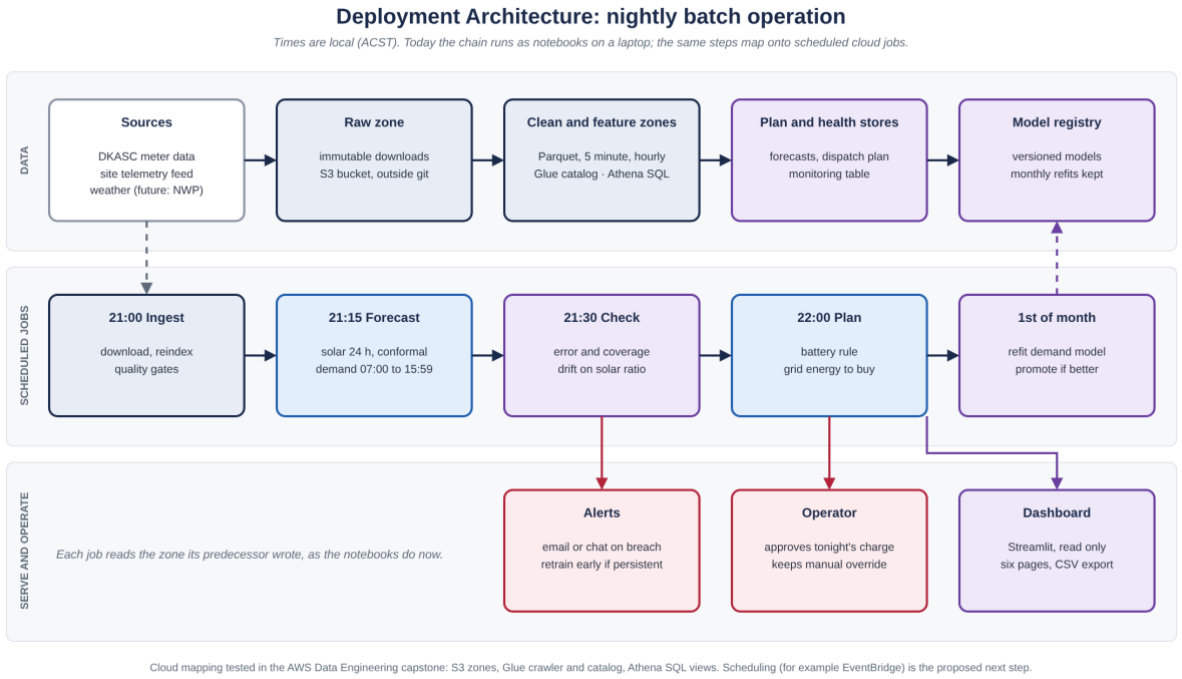


Figure 13. Deployment architecture. Today, the chain is notebooks; the data zones are on top of S3; the Glue catalog and Athena are on top of them, as experimented during the AWS Data Engineering capstone; the next proposed step is the use of a scheduler..

Failure or risk	Detection	Response already built or defined
Solar forecast missing for some hours	Gap check at ingest	Fall back to the same hour yesterday (used for 80 hours in the simulation)
Forecast quality degrades	Daily error and coverage limits	Alert; keep the current model; early refit if the alert persists
Site configuration changes	Drift check on the site solar ratio	The precedent from February 2024 – reset the reference and refit scale factor.
Demand regime changes	Monthly refit, retraining comparison	Monthly refit; triggered refit as backup

Failure or risk	Detection	Response already built or defined
Tariff differs from the assumption	Before go live, configuration review will be done.	Prices are parameters; rule turns to solar when the price is flat.
Dashboard unavailable	Process check	Plan is written to the store and can be exported as CSV
Wrong recommendation acted on	Operator review	Only for decision support, the operator approves and can override

Table 13. Operational risks and responses.

Packaging and reproducibility. The dashboard comes with a requirements file and theme file and requires 1 install command and 1 run command. The repository path is hard coded into the notebooks 1st cell, this is a known limitation described in the README which gets executed by run_pipeline.py, which runs notebooks 01 through 08 and checks that all 39 expected outputs (stores, result tables and 18 figures) exist and were rewritten by the run (SCRUM-51). The zone pattern that Figure 12 suggests was used in practice: the cloud pathway was implemented during the AWS Data Engineering capstone project where data were stored in S3, indexed with Glue crawler and accessed with Athena views.

What deployment still requires. A live data feed (the DKASC portal is a periodic download, which is now terminated on 22 August 2025), a scheduler and the site's actual tariff. All of these are NOT modelling problems; they are steps in integration between a validated decision support system and an operating system.

11. Governance, ethics, privacy and security

Decision support, not control. The system's power is intentionally restricted because a bad choice of battery costs money. It suggests to buy overnight, and it explains the reasoning (and the uncertainty), and the operator gives an approval (and can overrule). The dashboard is read-only and does not control equipment. Each save is presented side-by-side with the result of a perfect save and the result of the recorded operation, so that the performance is always provided in context and weak periods (Figure 8) are not concealed.

Privacy. Data are measurements taken at a public demonstration facility and a co located weather station – there is no personal information. Even with site demand data, it is possible to determine if a facility is in use. For a private operator, the same pipeline would require access control by zone (which the raw, clean and feature separation already provides) and the dashboard would be behind authentication (not on a local machine).

Security. The ignore rules also ignore environment, key and certificate files, and no credentials are stored in the repository or dashboard. In the AWS capstone, temporary lab credentials were only entered on the lab terminal and not into the repository or reports. The capstone also demonstrated least privilege: a deny was applied to one service which prevented a task being run, the event was reported back to the lecturer, and no effort was made to circumvent the control. No shared keys would be used on any cloud deployment, just roles that are used on each job.

Data licensing and culture. DKASC is credited in all reports, in the dashboard and in README, and raw data are not redistributed. As in Assessment 2, the prior working undertaken on this site does not involve the reuse of First Nations seasonal knowledge as this is cultural intellectual property and must be engaged with Traditional Owners.

Process governance. Changes were planned in Jira (Sprint 3 for the A3 demonstration, Sprint 4 for this report), and each change was merged into the main branch via a pull request that was reviewed and had the issue key in the commit message. The risk register has been updated, with G6 (narrow intervals) closed, February 2024 site change added as T12 realised and handled, new risks added to the register – demand pattern change (T13), tariff mismatch (T14), simulated savings overstating real

(T15), operator following a wrong recommendation (G8), and stale data (T5) remains open. Generative AI was used to draft code and report text for A3 and A4, as part of the same boundary that was disclosed in Assessment 2: the team outlined the work and ran each notebook, checking each number against its output and writing individual reflections (full disclosure is saved in Supporting_Documents).

12. Limitations and future improvements

Limitation	Effect	Future improvement
Illustrative tariff	Dollar values indicate scale, not a contract savings; valid only for time of use	Apply the site tariff (including demand charge) set by the site.
Simulation, not live operation	Takes it for granted that the battery is to follow the plan exactly	Shadow mode trial: execute the plan alongside with the current operation and compare the results
No weather forecast input	24 h solar forecast: based on persistence and season, cloudy days are neglected (27 May 2025)	Include numerical weather prediction irradiance and cloud cover.
Rule based decision	Captures 70% of the perfect foresight benefit; plans one night at a time	Use linear programming or model predictive control to optimise the next 24-48 hours
Data end on 22 August 2025	No live validation; models not tested on newer seasons	Live feed and scheduled ingestion
Single scale factor from Master Meter 1 to site	Site forecast inherits meter error plus 3.4 kW scaling error	Train on site solar, once sufficient post change history exists.
Marginal, not conditional, coverage	Overall, 80% of the time, the interval is correct, but on unusual months (August 2025 alert) it can be false.	Condition calibration on season or forecast level
Battery degradation ignored	Extra cycles have a hidden cost	Add a per cycle wear cost to the decision
Demand model has no occupancy or event data	The impact of public holiday and events remains difficult.	Include a holiday calendar and site schedule.

Table 14. Limitations, their effect on the results, and the improvement each points to.

The order of future work is based on the evidence in this report. The remaining gap to perfect foresight is addressed by the weather forecast, the optimiser targets the gap between the simulated and the real tariff by introducing a shadow mode trial, and the failure cases the monitoring shed light on are resolved by conditional calibration and a holiday calendar.

13. References

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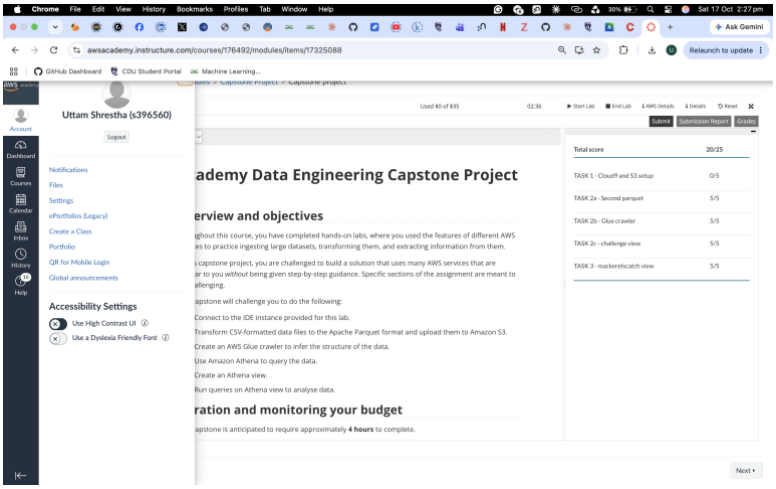
Section 4. Individual Work and Contribution Summary

1. Student Information

Name	Uttam Shrestha	**Student ID**	s396560
Role	Project Lead	**GitHub**	@Uttam1618

2. Individual Capstone Project Completion Evidence

AWS Academy Data Engineering · Capstone project



AWS Academy Data Engineering Capstone Project · total score 20/25 (Task 1 0/5; Tasks 2a, 2b, 2c and 3 5/5 each) · captured 17 October 2026, 2:27 pm · Task 1 blocked by an explicit deny on Cloud9, reported to the lecturer

3. Individual Project Contribution

- **Calibration and battery planning (A3).** Successfully saved all 113,677 out of sample forecasts, created rolling conformal calibration (daylight coverage 0.706 to 0.800), developed the battery model using the site telemetry and the dispatch simulation for the night (22.3% saving): notebook 06, PR #2; SCRUM-44, 45, 46, 47.
- **Monitoring, drift and retraining (A4).** Daily error and coverage checks with fixed limits, drift check that detects the February 2024 site change, comparison of three retraining strategies: notebook 08, PR #6; SCRUM-63.
- **Decision support dashboard (A4).** Streamlit app with six pages and the battery rule rerun live, checked against notebooks 06 and 07: dashboard/, PR #7; SCRUM-64.
- **Project management and reporting.** Sprint 3 and 4 planning and Jira upkeep (SCRUM-48, SCRUM-62); A3 deck and live demo (SCRUM-54, SCRUM-58); report Sections 2 and 3 and new architecture diagrams 09 and 10 (SCRUM-68); reviewed PRs #3, #4 and #5.
- **AWS Data Engineering Capstone.** S3 buckets, Parquet conversion, Glue crawler and Athena views; Task 1 blocked by an explicit deny on Cloud9, reported to the lecturer (SCRUM-70).

4. Brief Professional Reflection

In Assessment 2 my biggest contribution was to make our solar forecast a decision. Calibrated the prediction intervals, created a battery model using 2024 site telemetry, simulated a nightly charging plan and added monitoring, drift detection and the Streamlit dashboard. The most difficult issue was its sign convention was not documented and there was a telemetry configuration change in February 2024. I ran all four sign combinations against the energy balance and only scaled the data on the post-change data which we checked with a drift test later. A comparison with perfect foresight also revealed that demand, not solar, was our weak spot when measured in dollars, forcing the Assessment

4 effort to focus on demand. All of the members' work was integrated into the same system as I planned Sprints 3 & 4, reviewed all pull requests, and created the report.

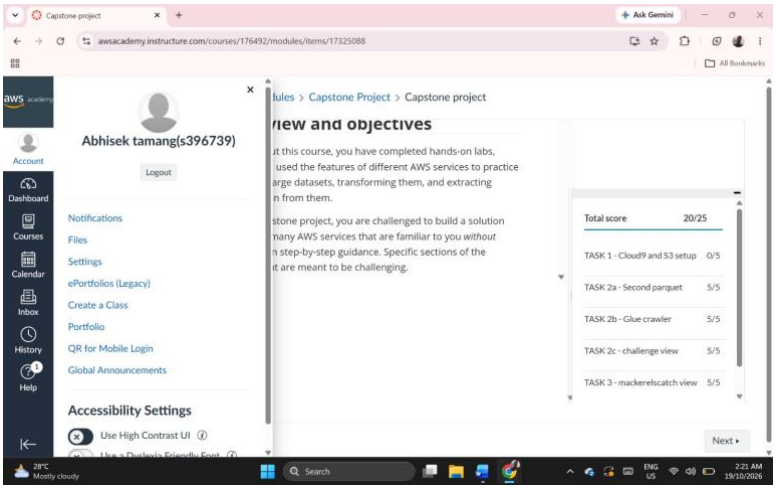
Section 4. Individual Work and Contribution Summary

1. Student Information

Name	Abhisek Tamang	**Student ID**	S396739
Role	Data Engineer	**GitHub**	@Abhisek000

2. Individual Capstone Project Completion Evidence

AWS Academy Data Engineering · Capstone project



AWS Academy Data Engineering Capstone Project · total score 20/25 (Task 1 0/5; Tasks 2a, 2b, 2c and 3 5/5 each) · captured 19 October 2026, 2:21 am

3. Individual Project Contribution

- **Architecture (A3).** Updated system architecture diagram 08 as built at Assessment 3: PR #4; SCRUM-49.
- **Repository documentation (A3).** README updated for A3 and data files kept out of git: PR #4; SCRUM-50.
- **A3 presentation.** Presented and recorded slides 1 to 5 (overview, objectives, architecture, A2 data recap): SCRUM-56.
- **Final architecture (A4).** As built and deployment architecture diagrams 09 and 10 (draw.io and PNG): PR #10; SCRUM-65.
- **Governance (A4).** Assessment 4 risk review (interval risk closed, five new risks) and AI use disclosure for A3 and A4: PR #10; SCRUM-66.
- **AWS Data Engineering Capstone:** SCRUM-71.

4. Brief Professional Reflection

The most notable piece of work since Assessment 2 which documented the changes in the system as it grew was my most significant. Changing the architecture diagram for Assessment 3 to include new calibration and battery layers, updated the README, and ensured that large data files were not added to git. Assessment 4 involved finalising the architecture and deployment views and the Governance documents (disclosure of AI use, risk register). The biggest problem I had was to keep the diagrams as accurate as possible while the notebooks were still in flux, which I handled by verifying each part of the diagram with the committed notebooks and outputs before exporting. I also shared the first few slides for our Assessment 3 video about the goals and structure of the video. My contribution was to produce a map of the system and its controls - this helped the technical work to be followed and verified with the marker.

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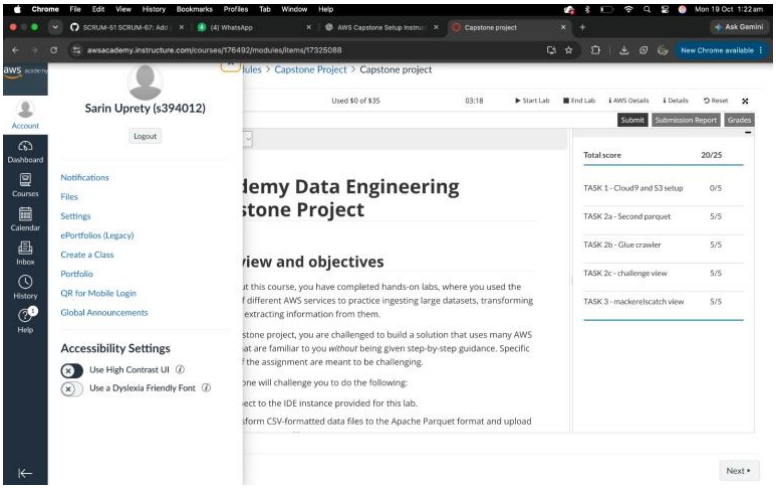
Section 4. Individual Work and Contribution Summary

1. Student Information

Name	Sarin Uprety	**Student ID**	S394012
Role	Data and Feature Engineer	**GitHub**	@Sarin751

2. Individual Capstone Project Completion Evidence

AWS Academy Data Engineering · Capstone project



AWS Academy Data Engineering Capstone Project · total score 20/25 (Task 1 0/5; Tasks 2a, 2b, 2c and 3 5/5 each) · captured 19 October 2026, 1:22 am

3. Individual Project Contribution

- **Workflow (A3).** Workflow diagram from data to decision and team process: PR #3; SCRUM-52.
- **Code review (A3).** Reviewed and approved the A3 analysis pull request: PR #2; SCRUM-53.
- **A3 presentation.** Presented and recorded slides 6 to 9 (workflow, progress, A2 model recap, calibration): SCRUM-57.
- **Reproducibility (A4).** run_pipeline.py, which runs notebooks 01 to 08 and checks all 39 expected outputs: PR #11; SCRUM-51.
- **Documentation (A4).** README updated for the final system (notebooks 07 and 08, reproducibility check, A4 results): PR #11; SCRUM-67.
- **AWS Data Engineering Capstone:** SCRUM-72.

4. Brief Professional Reflection

Since Assessment 2 my main contribution was making the project easy to follow and to reproduce. Designed workflow diagram that depicts the journey from raw data to the nightly battery decision and how our team got work through Jira and pull requests. I examined and accepted the pull request for Assessment 3, checked the calibration and dispatch output and gave the presentation for slides 6-9 of our video. In Assessment 4 I did work on the pipeline runner, the reproducibility check and the README for the final system. The big problem was the notebooks each needed files from the preceding notebooks, and each had an expected folder structure, and I recorded the sequence of running the notebooks and the output expected from each notebook so someone else can redo the entire chain. This helped the project by making it possible for other people to repeat, and trust, the process of putting notebooks together.

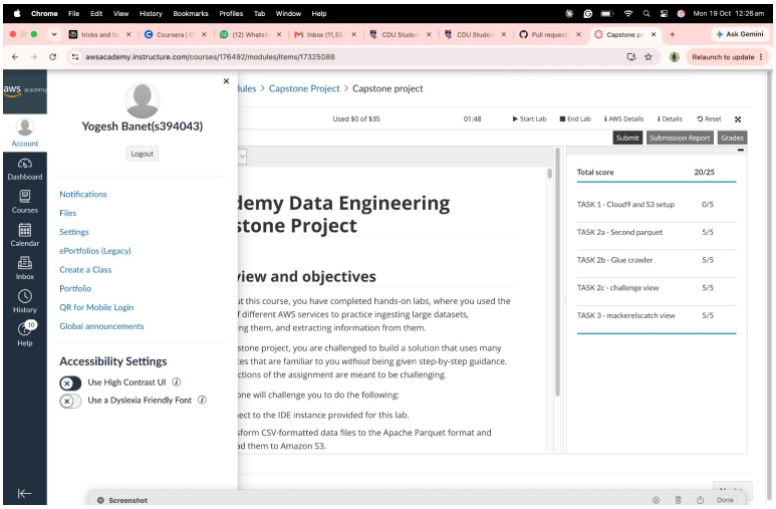
Section 4. Individual Work and Contribution Summary

1. Student Information

Name	Yogesh Basnet	**Student ID**	S394043
Role	Modelling and Visualisation Lead	**GitHub**	@viperx-ux

2. Individual Capstone Project Completion Evidence

AWS Academy Data Engineering · Capstone project



AWS Academy Data Engineering Capstone Project · total score 20/25 (Task 1 0/5; Tasks 2a, 2b, 2c and 3 5/5 each) · captured 19 October 2026, 12:26 am

3. Individual Project Contribution

- **A3 video and deck.** Merged, checked and uploaded the group video; committed the deck and individual PDFs: PR #5; SCRUM-55, SCRUM-61.
- **A3 presentation.** Presented and recorded slides 13 to 16 (challenges, limits, future work): SCRUM-59.
- **Demand forecasting (A4).** Five demand approaches compared walk forward and the battery plan rerun with each (gradient boosting 24.9 kW against 54.5 kW; share of perfect foresight 39% to 70%): notebook 07, PR #8; SCRUM-60.
- **Quality review (A4).** Final report review against the marking rubric: SCRUM-69.
- **AWS Data Engineering Capstone:** SCRUM-73.

4. Brief Professional Reflection

The most important work since Assessment 2 was the demand forecast comparison I did in notebook 07. After doing analysis in Assessment 3, I found that it was not solar, but demand that limited the battery plan, therefore I compared five approaches with monthly walk forward evaluation. The planning hour error decreased from 54.5 kW to 24.9 kW with gradient boosting and the perfect foresight benefit increased from 39% to 70% in the battery plan. The biggest problem was that the old seven-day mean was poorly performing at the weekends (when the site is closing almost entirely), adding weekday lag and recent lag fixed this. Assessment 3: Committed video, final slides on challenges and future work. The final report was also checked against the rubric in Assessment 4. My work reinforced and bolstered the weakest link in the system and the overall value of the result was increased.